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ANALYSIS OF MECHANICAL STRUCTURES
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Nomenclature

Symbol	Name	Units of measurement
ρ	Volumetric Density	$[Kg/m^3]$
E	Young's modulus	Pa
$\mathbf{E}$	Young's modulus tensor for orthotropic materials	Pa
ν	Poisson's ratio	-
μ	Neo-Hookean	$[Pa]$
κ	Neo-Hookean	$[Pa]$
J	Jacobian transformation tensor	-
K	Stiffness matrix (elemental or global)	N/m
M	Mass matrix (elemental or global)	Kg
a	Rayleigh method's mass coefficient	s
b	Rayleigh method's stiffness coefficient	$1/s$
C	Damping matrix	Damping matrix
t	Thickness	m
t	Time	s
U	Displacement Vector	m
$\vec{\Phi}$	Mode	$[-]$
P	1 st Piola – Kirchoff Stress tensor	$[Pa]$
$[S]$	2 nd Piola – Kirchoff Stress tensor	$[Pa]$
$[\epsilon_{GL}]$	Green- Lagrange Strain tensor	$[-]$
D	Material State tensor	$[Pa]$

Chapter I

1.1 Computational Framework and System Architecture

Inspired by the architecture of industrial finite element analysis (FEA) software, this work develops a modular standalone computational library for finite element simulations. The primary objective is to provide an object-oriented and extensible platform suitable for multiphysics applications and interdisciplinary engineering studies. By adopting a decoupled software structure, the framework enables complex engineering problems to be defined and solved in a flexible and efficient manner.

Furthermore, particular emphasis was placed to abstracting input parameters into minimalist configurations. This minimizes the steep learning curve often encountered by students and young researchers when interacting with dense, commercial software ecosystems. Crucially, the library's application programming (API) is engineered to allow seamless, user-defined algorithmic extensions of the current algorithm for advanced structural variants.

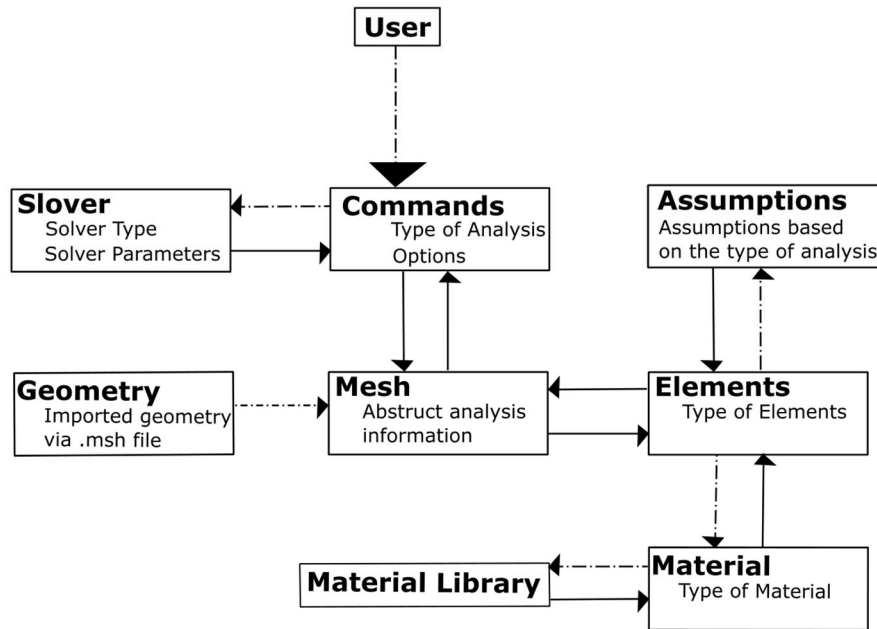


Figure 1. Information Diagram of the custom program

The above diagram depicts the algorithmic interaction between the core entities of the computational library. The dashed arrows denote user-level interactions, through which the default parameters of specific entities may be modified. In contrast, the solid arrows indicate the flow of information between the core entities during the assembly and solution of an analysis. The user initializes an analysis using predefined execution commands, instantiating an object derived from an abstract base Project class. Specific analysis suites - such as dynamic modal extraction or geometric non-linear formulations - inherit from this parent class, expanding the variable space to handle specialized boundary conditions and solvers.

Each project is associated with a set of assumptions, irrespective of the analysis type. These assumptions are represented as a list of predefined keywords corresponding to an enumerated assumption class. Each analysis activates only the assumptions relevant to its formulation, while the remaining ones are ignored. In this way, the software structure remains simple and well organized, even after a large number of consecutive analyses. The pointer for the list of assumptions is stored in every element of the corresponding mesh, giving the possibility of different assumptions in different partitions of the same mesh.

The mesh also stores the boundary conditions associated with the problem. In the current work the only types of boundary conditions that have been pre-defined are:

- ***SlidingX, SlidingY***: Constrains a node such that its field variable may vary only along the specified direction.
- ***Fixed***: Constrains all degrees of freedom of a node.
- ***Displacement***: Applies a displacement u along a specific global axis. A vector-valued displacement boundary condition may also be applied through the corresponding command.
- ***Force***: Applies a nodal force vector F at a specified mesh node.

Despite their simplicity, assigning boundary conditions individually at each node is inefficient in practical applications. Therefore, the command set also supports edge-based assignment, whereby a boundary condition may be applied automatically to all nodes lying on a straight boundary segment defined by two end points.

1.2 Mesh and Continuum Discretization

The mesh acts as the critical data abstraction bridge connecting the numerical solvers to the fundamental continuum mechanics governing the domain. To support multi-disciplinary configurations, the spatial grid is decoupled from the specific problem physics. Nodes are structured to handle arbitrary degrees of freedom (DoFs), granting the framework flexibility across diverse physics fields.

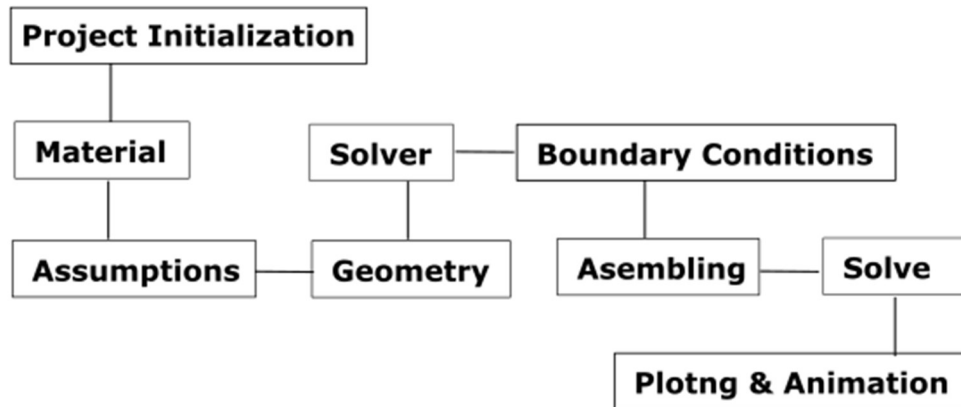


Figure 2. Workflow of an arbitrary project

The physical mesh topology is constructed by a dedicated Mesher. Meshers are entities that use user-defined geometries, that outline the structures and based on their parameters construct a corresponding mesh. This algorithm parse discrete node coordinates and elemental connectivity arrays generated from external “.msh” geometry files. Currently, at this stage of the project, it currently provides two methods to generate a mesh from a given geometry:

- ***Rectangular Division 2D Mesher***: It needs the edge divisions of the rectangular 3d Shape.
- ***Uniform Density 2D Mesher***: It needs constant mesh density.

All baseline internal parameters default strictly to the international System of Units. However, for customized configurations, the library enforces dimensional consistency across all governing equations. Last but not least, the solvers for each analysis will be discussed in every following part of this report.

1.3 Material Constitutive Formulations

One of the most crucial parts of the simulation is the material assignment and the subsystem (library) that stores the materials and their properties. Material definitions are handled via a generalized abstract class engineered to accommodate both linear elastic and non-linear material laws (reference to chapter IV). Derived from these, every type of material then has its own subclass utilizing its parameters. Predefined material regimes are stored within a structured JSON database that maps distinct identifies to their physical constants:

Table 1. Material Type table

Material Type	Primary Field Properties	Description
Material	ρ , Type	Generalized base class definition; Abstract placeholder requires explicit subclass instantiation.
Isotropic	E, ν	Classical linear elastic material demonstrating orientation - independent mechanical properties.
Orthotropic	[E], ν	A family of materials, which exhibits mutually orthogonal planes of elastic symmetry.
Neo-Hookean	μ , κ	Isotropic hyperelastic formulation governed by an objective strain energy density function W .

Every user has the possibility to define and store new materials, and they are saved also for other projects that use the same library. In addition, there is also a temporary defining of a material inside each analysis.

Chapter II

Part I

2.1 Quadrilateral 2D elements, theoretical background

The 4-node quadrilateral (Quad4) shell and plate elements represent a significant advancement over simpler triangular elements, making them the industry standard for modeling thin-walled structures. In a standard Cartesian formulation, the unknown displacement field within a four-node element is typically approximated using a bilinear polynomial equation (containing terms up to the first order for each variable individually, i.e., 1, x, y, xy).

$$u(x, y, t) = a_0(t) + a_1(t)x + a_2(t)y + a_3(t)xy$$

However, restricting the element formulation to a standard Cartesian coordinate system introduces severe limitations. Most notably, the inter-element C^0 continuity of the displacement field along the boundaries is compromised if the element edges are not perfectly parallel to the global Cartesian axes. To overcome this geometric dependency and eliminate the rigid requirement for rectangular meshes, modern finite element formulations employ the isoparametric concept. By mapping the physically distorted element in the global Cartesian domain Ω_{xy} onto a regular, bi-unit parent domain Ω_{rs} where $(r, s) \in [-1, 1]^2$ or $(r, s) \in [0, 1]^2$ (Langrange isoparametric elements) via a coordinate transformation, C^0 -continuity is strictly preserved across arbitrary quadrilateral geometries. This mapping ensures that compatibility is maintained along inter-element boundaries, independent of the displacements of the nodes of the element in the physical system.

$$x(r, s) = \sum_{i=1}^N N_i(r, s)x_i \text{ and } y(r, s) = \sum_{i=1}^N N_i(r, s)y_i$$

At the isoparametric element Quad4, the unknown displacement field is approximated by the following bilinear shape functions, defined for the parent coordinates $r, s \in [-1, 1]$:

$$N_1(r, s) = \frac{1}{4}(1 - r)(1 - s)$$

$$N_2(r, s) = \frac{1}{4}(1 + r)(1 - s)$$

$$N_3(r, s) = \frac{1}{4}(1 + r)(1 + s)$$

$$N_4(r, s) = \frac{1}{4}(1 - r)(1 + s)$$

The implementation of the isoparametric Quad4 elements requires the computation of spatial derivatives, because the equations must ultimately be evaluated in the physical coordinate system x, y rather than the parent domain r, s . To establish the geometric connection between the two coordinate systems, the following mapping transformation was introduced:

$$(r, s) \mapsto (x(r, s), y(r, s))$$

By invoking the chain rule of differentiation, the relation between the parametric gradient and the physical gradient is governed by the Jacobian operator, J , defined as follows:

$$\begin{bmatrix} \frac{\partial}{\partial r} \\ \frac{\partial}{\partial s} \end{bmatrix} = J \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix}$$

Expanding this relation reveals the components of the 2×2 Jacobian matrix:

$$J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial y}{\partial r} \\ \frac{\partial x}{\partial s} & \frac{\partial y}{\partial s} \end{bmatrix}$$

Utilizing the isoparametric assumption, the physical coordinates within the element are interpolated using the same bilinear shape functions and the discrete nodal coordinates (x_i, y_i) : Substituting these spatial interpolations into the Jacobian matrix yields its explicit, computable form based entirely on the geometry of the element nodes.

$$J = \begin{bmatrix} N_{1,r} & N_{2,r} & N_{3,r} & N_{4,r} \\ N_{1,s} & N_{2,s} & N_{3,s} & N_{4,s} \end{bmatrix} \cdot \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix}$$

Transformation of stiffness and mass matrices

In a conventional, non-isometric finite element formulation, the element stiffness matrix K_e and the consistent mass matrix M_e at every element are defined via integrals over the physical volume V_e of the element:

$$[K]_e = \int_{V_e} B_{xyz}^T D B_{xyz} dV$$

$$[M]_e = \rho \int_{V_e} N^T N dV$$

Where B_{xyz} contains the derivatives with respect to the physical coordinates x, y, z , and for a planar element of uniform thickness t , $dV = t \cdot dxdy$.

When transitioning to the isoparametric Quad4 element, these conventional equations undergo a fundamental transformation in two areas: the differential volume element and evaluation of the shape function derivatives. As established by the change of variables theorem in calculus, integrating over the distorted physical coordinates x, y is mapped to the flat, bi-unit parametric domain $(r, s) \in [-1, 1]^2$. The differential physical area relates to the parametric area through the determinant of the Jacobian matrix:

$$dxdy = \det(J) drds$$

This substitutes a highly complex boundary integration in physical space with fixed limits of integration to $-1, 1$. The conventional matrix relies on physical spatial derivatives $(\partial N_i / \partial x, \partial N_i / \partial y)$. In the isometric system, these are replaced by multiplying the parametric derivatives by the inverse Jacobian operator J^{-1} :

$$\begin{bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \end{bmatrix} = J(r, s)^{-1} \begin{bmatrix} \frac{\partial N_i}{\partial r} \\ \frac{\partial N_i}{\partial s} \end{bmatrix}$$

Consequently, the strain-displacement matrix is no longer constant, but becomes a complex function of the parametric coordinates, denoted as $B(r, s)$.

By applying these two structural changes, the conventional integrals are evaluated in the parent coordinate system:

$$[K]_e = t \int_{[-1,1] \times [-1,1]} B(r, s)^T \cdot D \cdot B(r, s) \cdot \det(J(r, s)) drds$$

$$[M]_e = \rho t \int_{[-1,1] \times [-1,1]} N(r, s)^T \cdot N(r, s) \cdot \det(J(r, s)) drds$$

Because these matrices are dependent deeply on the r and s , finding an analytical formula for these equations is nearly impossible. This operation hurdle is precisely what requires the transition from continuous integration to discrete Gauss - Legendre Quadrature, where the integrals are evaluated as weighted sum at specific sampling points (r_i, s_j) :

$$[K]_e \simeq t \sum_{i=1}^n \sum_{j=1}^n W_i W_j B(r_i, s_j)^T \cdot D \cdot B(r_i, s_j) \cdot \det(J(r_i, s_j))$$

$$[M]_e \simeq t\rho \sum_{i=1}^n \sum_{j=1}^n W_i W_j N(r_i, s_j)^T \cdot N(r_i, s_j) \cdot \det(J(r_i, s_j))$$

Lumped Mass Matrix

The lumped mass formulation is a kinematic approximation that idealizes the continuous mass distribution of the element as a discrete set of concentrated point masses localized strictly at the nodal positions. This usual kinematic assumption decouples the inertia interactions between different degrees of freedom, effectively eliminating all off-diagonal coupling terms.

$$[M]_e^{lumped} = \text{diag}(m_1, m_1, m_2, m_2, \dots, m_N, m_N)$$

Where m_i represents the discrete mass fraction allocated to node i .

To compute these nodal mass fractions while strictly preserving the total conservation of elemental mass, the row-sum technique is applied. This algebraic method evaluates the spatial integral of the consistent mass matrix row entries, collapsing the formulation down to:

$$m_i = apt \int_{[-1,1] \times [-1,1]} N_i(r, s) \cdot \det(J(r, s)) dr ds$$

The dimensionless scaling factor a is a correction factor enforced to ensure that the sum of the nodal masses identically matches the total physical mass of the element. Therefore, it is calculated by the following formula:

$$a = \frac{M_{total}}{\sum m_i}$$

For uniform, undistorted rectangular geometries, the shape function symmetry naturally yields an equitable distribution where $a = 1$ and $m_i = 1/4 * M_{total}$. For distorted quadrilateral geometries, a scales the coefficients to prevent numerical mass loss or creation during discretization.

2.2 Isoparametric Limitations and Anomalies

The mathematical Bounds of the Isoparametric Mapping

For the coordinate transformation from the bi-unit parent domain Ω_{rs} to the physical domain Ω_{xy} to be mathematically valid, the mapping must be a one-to-one transformation (bijective) and invertible. This requirement hinges entirely on the determinant of the Jacobian matrix, $\det(J(r, s))$.

To ensure a valid, non-singular, mapping the following condition must be strictly satisfied everywhere within the element, and subsequently everywhere inside the mesh.

$$\det(J(r, s)) > 0, \forall (r, s) \in [-1, 1]^2$$

If this equation is violated, the finite element formulation yields, and the solution cannot be continued. In the first case, the determinant of the Jacobian being zero, there is a matrix inversion failure. Specifically, in Quad4 elements, this situation occurs when three nodes are collinear with each other, completely flattening out one of the corners. Thus, the element locally loses a geometric dimension, as a result it is not able to define the surface at the exact vertices and the element area shrinks to zero. In the second case that the determinant has a negative

value the shape of the element has become concave, and especially in the quadrilateral elements is the characteristic arrowhead shape.

Numerical pathology of Hourglass Modes (Zero-Energy Modes)

For a finite element discretization to remain stable, the rank of the elemental stiffness matrix $[K_e]$ must satisfy the algebraic constrain determined by the total elemental degrees of freedom (n_{dof}) and the number of rigid body modes (n_{rbm}):

$$rank([K_e]) = n_{dof} - n_{rbm}$$

When under-integrated (reduced-integration) quadrature rules are deployed- such as a single Gauss point(1x1) for a bilinear 4-node quadrilateral element - the matrix B is sampled exclusively at the element centroid. Consequently, the dimension of the null space of $[K_e]$ expands beyond the physically defined rigid body modes. This phenomenon induces a rank deficiency, giving rise to spurious, kinematically admissible displacement vectors u_h that satisfy:

$$[K_e] \cdot u_h = 0, \text{ where } u_h \notin U_{rbm}$$

Because these deformation modes elicit zero strain at the Gauss points ($B \cdot u_h = 0$), the internal virtual work and the corresponding strain energy vanish ($U = 0$). Hence, they propagate through the mesh with zero resistance.

In a physical context, these modes emerge primarily under localized bending gradients. The alternate extension and contraction of element boundaries cancel out entirely at the central Gauss point, rendering the quadrature point “blind” to the macro scale deformation.

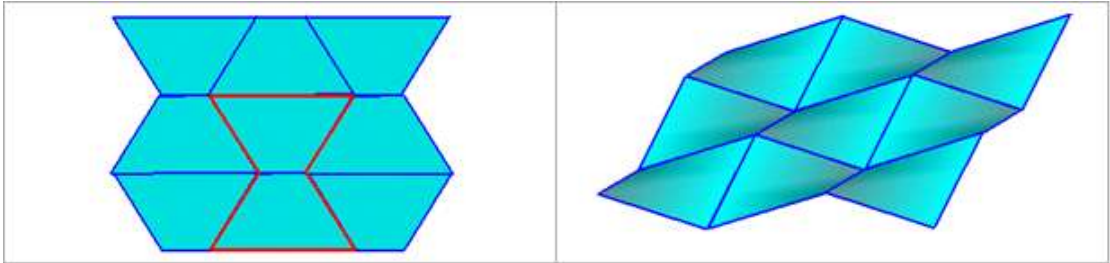


Figure 3. Hourglass modes

The illustration above demonstrates this manifestation within a continuous mesh. Since no macro-scale strain energy opposes this deformation, the unconstrained displacement fields propagate unchecked across adjacent boundaries, resulting in a characteristic periodicity, non-distortion, leading to an un-physical reduction in global structural stiffness and eventual divergence of the non-linear algebraic solver.

To retain the computational efficiency of reduced integration while eliminating rank deficiencies, specialized numerical remedies must be enforced. The most straight forward method is the full Gauss quadrature evaluation of the B matrix, but as discussed earlier this direction is able to cause shear locking inside the mesh. Following, *Perturbation Stiffness* is one of the simplest methods for averting the “Hourglassing”. This method introduces an orthogonal hourglass base vector γ to explicitly penalize the non-physical displacement fields (u_h). Despite its simplicity, this method required empirical tuning of scaling parameters to prevent excessive mesh stiffening. In addition, the method “Assumed Strain Method” modifies the dilatational or volumetric components of the B matrix independent of the deviatoric

components. The above solution is highly effective at preventing volumetric locking without introducing zero-energy modes.

To guarantee that numerical stabilization does not violate thermodynamic admissibility or artificially distort the structural response, the global energy balance must be strictly monitored. The artificial energy dissipated or stored by the stabilization scheme ($E_{Hourglass}$) must remain a negligible fraction of the total internal strain energy ($E_{internal}$):

$$\frac{E_{Hourglass}}{E_{internal}} \leq 0.05$$

Exceeding this 5% threshold signifies that the stabilization forces are contributing non-trivial and non-physical energy to the system, thereby invalidating stress and displacement.

Shear Locking (Parasitic Shear)

Shear locking, or known as “*parasitic shear*”, is a well-documented kinematic deficiency, inherent to fully integrated, low-order isoparametric elements. The most representative type of element is the 4-node quadrilateral (Quad4) shell and plate elements. This anomaly manifests as an artificial and severe overestimation of the element stiffness when it is subjected to dominated bending fields, causing the global finite mesh solution to converge to displacements that are orders of bigger shear deformation.

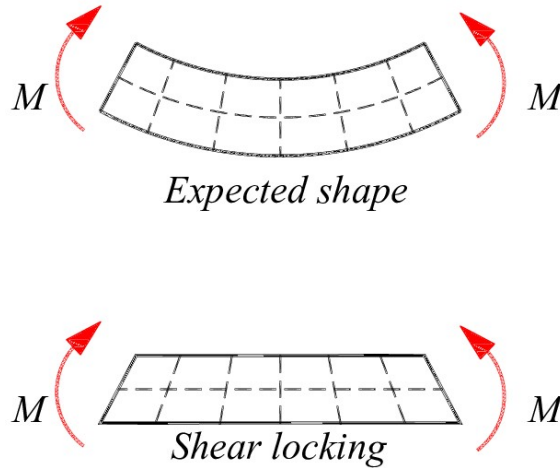


Figure 4. Shear Locking in a quadrilateral mesh

The main cause of the shear locking is the linear approximation of the bending stress field cannot depict correctly the real stress field. The Quad4 element boundaries are constraint to be straight lines. As a result, when the element is subjected to severe bending moments, the right angles to be acute and obtuse angles. The mathematical formulation interprets this change in angle as a severe deformation, even though the actual structural load is pure bending.

During the numerical integration (Gauss - Legendre Quadrature), at the points of sampling the formulation captures a false, parasitic shear strain. Because the shear modulus of thin-walled structures is highly dominant compared to bending stiffness, the algorithm places an enormous energy penalty on this fake shear strain. Consequently, the element rigidly “locks up”, spending its energy capacity resisting to non-existent shear forces rather than accommodating the true bending behavior.

There are some methodologies that can help resolve the above problem. If it is used low-order elements such as Quad4, or Hexa8 the first methodology is to increase the mesh density and also use reduce integration. The reduced integration method states that by computing the

Figure 5.

stiffness matrix with lesser Gauss nodes, the numerical errors from the approximation are minimized. However, if the extreme decrease of the Gauss nodes may lead to a phenomenon called spurious zero-energy modes. In addition, the use of incompatible mode elements can reduce the effect of shear locking in the final result. Finally, some algorithms, which are the industry standard for resolving this problem, are the Selective Reduced Integration (SRI) and Enhanced/ Assumed strain methods (EAS/ ANS).

In commercial finite element software like ANSYS, these isoparametric anomalies are managed natively through selectable element technologies embedded within lower-order elements. For structural thin-walled applications using the standard 4-node quadrilateral shell element (SHELL181), ANSYS defaults to the Assumed Natural Strain (ANS) method. By modifying the transverse shear strain shape functions using the Bathe-Dvorkin formulation, the software calculates shear strains at optimal edge midpoints rather than the standard Gauss points, completely filtering out parasitic shear locking without degrading element connectivity.

For 2D and 3D continuum elements (such as Plane182 and SOLID185), ANSYS provides specific key options (KEYOPT) to switch between the Enhanced Strain Formulation (EAS) - which injects internal degrees of freedom to counter pure bending inaccuracies -and the B-bar method (Selective Reduced Integration) to tackle separate volume locking issues. Furthermore, if reduced integration is chosen to maximize computational speed, ANSYS deploys an artificial hourglass stabilization energy check, allowing users to verify that the energy error remains safely under 5% to ensure a valid and stable simulation.

3 Structural Eigenvalue Analysis

3.1 Theoretical Background of Eigenvalue Analysis

The dynamic equilibrium of a continuous structural system discretized via Finite Element Method (FEM) is governed by the following system of second-order linear ordinary differential equations:

$$[M] \cdot \ddot{U}(t) + [C] \cdot \dot{U}(t) + [K] \cdot U(t) = F(t)$$

Where $[M], [C], [K] \in \mathbb{R}^{n \times n}$ represent the global mass, damping, and stiffness matrices respectively. The vector $U(t) \in \mathbb{R}^{n \times 1}$ is the global nodal displacement vector, and $F(t) \in \mathbb{R}^{n \times 1}$ is the vector of externally applied nodal forces. The parameter n denotes the total hardly unconstrained degrees of freedom (DoFs) in the system.

To determine the intrinsic dynamic characteristics of the 2D cantilever structure (natural frequencies and mode shapes), it is considered the case of free, undamped vibration. Consequently, the damping matrix C is set to be the zero matrix, $C = 0$ and the external force vector $F(t) = \vec{0}$, reducing the governing equation to a homogeneous second-order linear differential equation, as follows:

$$[M] \cdot \ddot{U}(t) + [K] \cdot U(t) = \vec{0}$$

We assume a harmonic, synchronous solution of the form:

$$U(t) = \vec{\Phi} \cdot e^{i\omega t}$$

Where $\Phi \in \mathbb{R}^{n \times 1}$ is the time independent vector of the nodal displacement amplitudes, representing the mode shape. Also, $\omega \in \mathbb{R}$ is the natural angular frequency of vibration.

Differentiating $U(t)$ twice with respect to time, yields the nodal acceleration vector. Then, after substituting $U(t)$ and $\ddot{U}(t)$ back into the undamped equation of motion:

$$-\omega^2[M] \cdot \vec{\Phi} e^{i\omega t} + [K] \cdot \vec{\Phi} = \vec{0} \Rightarrow ([K] - \omega^2[M]) \cdot \vec{\Phi} e^{i\omega t} = \vec{0}$$

Since $e^{i\omega t} \neq 0, \forall t \in \mathbb{R}^+$, the equation simplifies to the Generalized Algebraic Eigenvalue Problem:

$$([K] - \lambda[M]) \cdot \vec{\Phi} = \vec{0}$$

Where $\lambda = \omega^2$ represents the eigenvalue, and $\vec{\Phi}$ is the corresponding eigenvector.

The characteristic Equation and Eigenvalue Problem Solvability

For a non-trivial solution to exist ($\vec{\Phi} \neq \vec{0}$), the operator $[K] - \lambda[M]$ must be singular. According to the fundamental theorem of linear algebra, this requires its determinant to vanish:

$$p(\lambda) = \det([K] - \lambda[M]) = 0$$

Here, $p(\lambda)$ represents the characteristic polynomial of degree n . Since the global stiffness matrix K is positive-semidefinite (and becomes strictly positive-definite $[K] > 0$ after the imposition of essential boundary conditions at the fixed support) and the global matrix M is positive-definite $[K] > 0$, the roots of the polynomial yield n real, positive eigenvalues:

$$\lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_n$$

Properties of Eigenvectors: [M]-orthogonality

The eigenvectors $\vec{\Phi}_j$ satisfy the fundamental orthogonality conditions with respect to the mass and stiffness tensors due to the symmetry of K and M . For two distinct modes $j \neq k$:

$$\vec{\Phi}_j^T \cdot [M] \cdot \vec{\Phi}_k = 0 \text{ and } \vec{\Phi}_j^T \cdot [K] \cdot \vec{\Phi}_k = 0$$

Furthermore, because the fundamental governing equation $([K] - \lambda_i[M]) \cdot \vec{\Phi}_i = \vec{0}$ is a homogeneous linear system, the solution space for each eigenvalue is a subspace.

Consequently, if a vector $\vec{\Phi}_i$ is a solution, any scaled variant:

$$\vec{\Psi}_i = c\vec{\Phi}_i, \forall c \in \mathbb{R}^*$$

is also a mathematical valid eigenvector. This mathematical property directly underscores two critical physical and computational realities of structural dynamics. Firstly, the directional invariance of the mode shapes $\vec{\Phi}_i$, by setting $c = -1$ establishes that both $\vec{\Phi}_i$ and $-\vec{\Phi}_i$ are the identical solutions. Physically, this represents the harmonic nature of the structure's motion, which oscillates symmetrically back and forth between these two extreme deformed profiles over each period of vibration T_i . Furthermore, the eigenvalue problem only determines the shape of the structure, and not absolute physical magnitude of the deformation, making the above system practically indeterminate of amplitude.

In computational frameworks, eigenvectors are typically scaled via a mass-normalization, such as:

$$\vec{\Phi}_j^T \cdot [M] \cdot \vec{\Phi}_j = 1 \Rightarrow \vec{\Phi}_j^T \cdot [K] \cdot \vec{\Phi}_j = \omega_j^2$$

This mathematical property decouples the equations of motion in the modal space, establishing the theoretical prerequisite for the calculation of the Rayleigh damping matrix coefficient a, b using the first two target modes.

3.2 Results and Comparisons

The benchmark case considered in this section is a steel cantilever beam with dimensions 4 [m] $\times$ 1 [m] and a uniform thickness of 0.2 [m] across its length. The beam is fixed at its left end and discretized using the same mesh and material properties in both the custom program and ANSYS, so that any differences in the results can be attributed primarily to the numerical formulation rather than to modeling inconsistencies. During the setup of the analyses in ANSYS, all geometric nonlinearity options were disabled. Lastly, both the custom program and ANSYS were assigned the plane stress assumption.

At the end of the analysis, both programs exported four heatmaps representing the total nodal displacement of the beam. The custom program was then also run with the lumped-mass assumption, and the corresponding results are included in Appendix A.

Eigenvalues and modes comparison

This section validates the custom eigenvalue solver by comparing the computed natural frequencies and mode shapes of the cantilever beam with the corresponding results obtained from ANSYS. Such benchmarking provides a direct assessment of the accuracy of the implementation and highlights the influence of numerical features available in commercial finite element software, such as element technologies used to mitigate locking effects. The mode-shape plots for Modes 2 to 4 from both implementations are included in Appendix A.

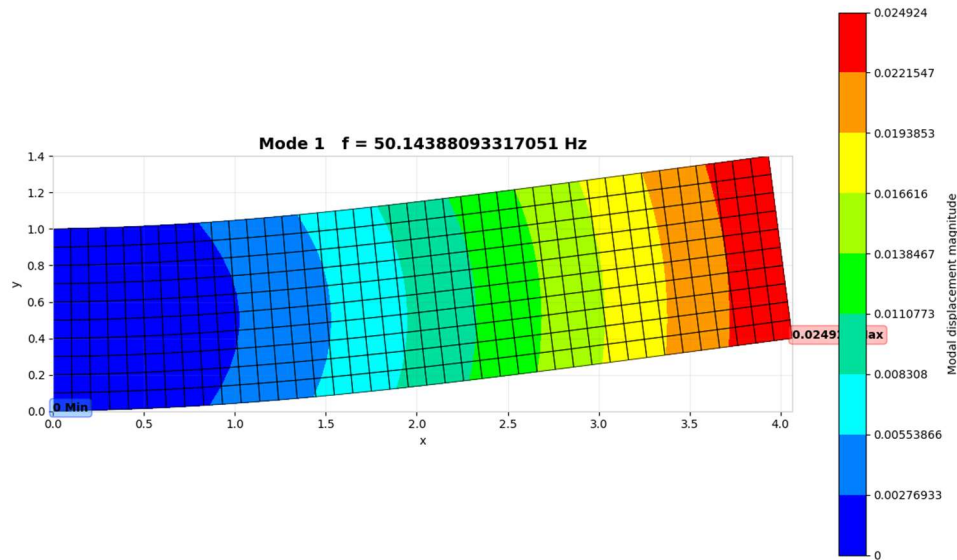


Figure 6. First mode depiction (Program)

Table 2. Frequency comparison between ANSYS and Program

Mode	Program Frequency [Hz]	ANSYS Frequency [Hz]	Deviation[%]
1	50.144	50.021	+0.25
2	255.128	254.6	+0.21
3	324.185	324.16	+0.01
4	587.47	586.32	+0.20

The frequency comparison in Table 2 shows that the custom implementation reproduces the ANSYS benchmark with very good accuracy for the first four modes. The smallest deviation is

observed for Mode 3, which is predominantly axial and therefore less sensitive to bending-related discretization effects. The slightly larger deviations in the bending-dominated modes are likely associated with the limitations of lower-order quadrilateral elements, particularly, as discussed earlier in Section 2.2, their sensitivity to shear locking and element formulation details. These differences remain small and do not affect the correct identification of the dominant modal characteristics of the structure.

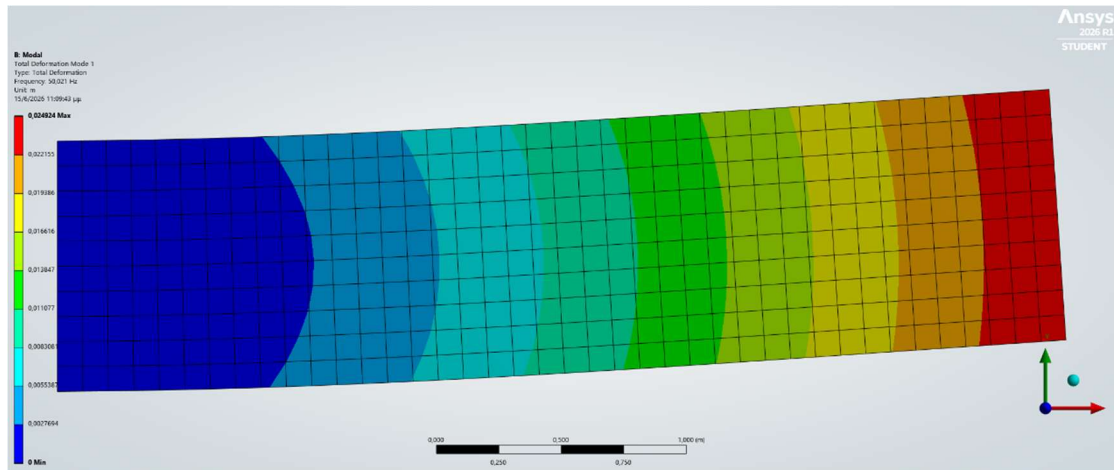


Figure 7. First mode depiction (ANSYS)

It is worth noting that some modes, particularly Modes 2 and 4, appear mirrored when the results of the two solvers are compared. Likewise, Mode 3 may appear as a tensile mode in one solution and as a compressive mode in the other. As explained in Section 3.1, this apparent discrepancy is purely mathematical and follows from the fact that eigenvectors are defined only to an arbitrary multiplicative constant, usually such as the eigenvector magnitude to be equal to one. In particular, if a vector is an eigenvector, then its negative represents the same mode shape and is associated with the same natural frequency. Therefore, the sign difference in the amplitude of the eigenvector has no physical significance and does not indicate disagreement between the two solvers.

Comparison between Lumped and Consistent Mass Approach

In structural dynamics, the choice between a lumped and a consistent mass matrix influences how inertia is represented in the eigenvalue problem. A consistent mass matrix uses the same interpolation functions as the displacement field and therefore provides a more accurate description of distributed inertia and modal coupling. By contrast, a lumped mass matrix concentrates the

elemental mass at the nodes, producing a diagonal matrix that is computationally efficient but less precise, especially for higher modes. As a result, the first few natural frequencies are often very similar for both approaches, whereas larger differences may appear in the higher-frequency range.

This distinction also affects subsequent transient analyses. A consistent mass matrix generally yields a smoother and more physically representative propagation of inertia throughout the mesh, whereas a lumped mass matrix is particularly attractive in explicit time integration because its diagonal form eliminates the need to solve a coupled linear system at each time step. The trade-off is that mass lumping may introduce local numerical artifacts, especially near points of concentrated loading, and may therefore affect the early-time response unless the mesh is sufficiently refined.

Table 3. Frequency comparison between mass assumption

Mode	Lumped Mass [Hz]	Consistent Mass [Hz]	Deviation [%]
1	50.122	50.144	$-4.4 \cdot 10^{-4} [\%]$
2	254.518	254.6	$-3.2 \cdot 10^{-4} [\%]$
3	324.14	324.16	$-6.2 \cdot 10^{-5} [\%]$
4	584.709	586.32	$-2.7 \cdot 10^{-3} [\%]$

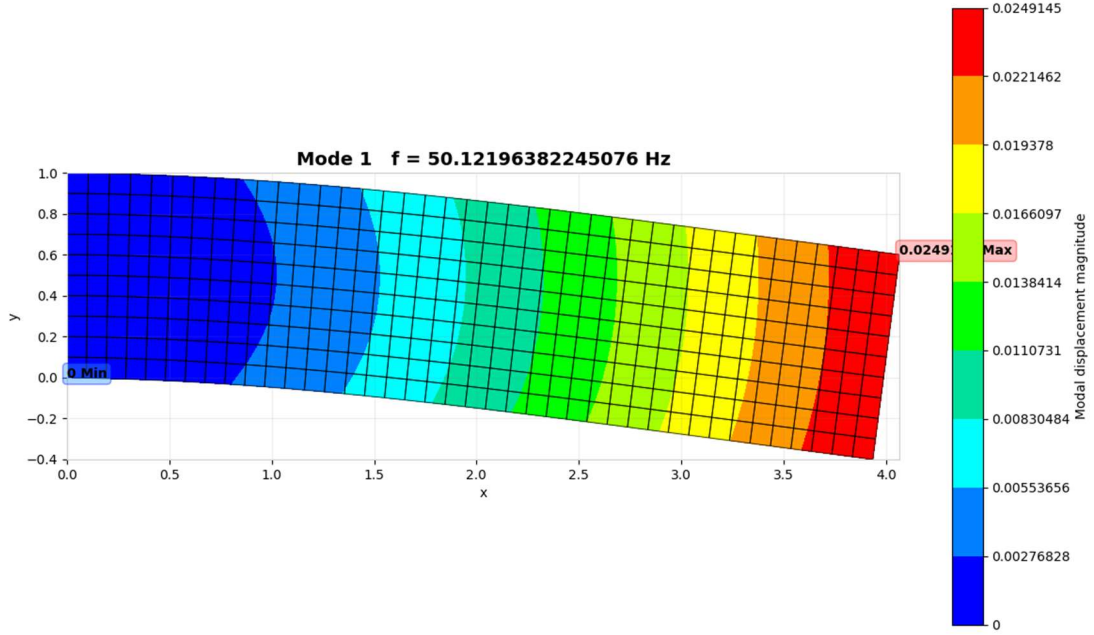


Figure 8. First mode, Lumped mass assumption (Program)

4 Transient Structural Analysis

4.1 Theoretical Background of Transient Analysis

When a structure is subjected to an explicit time-dependent external excitation, its behavior is governed by the full, linear second order system of differential equations:

$$[M] \cdot \ddot{U}(t) + [C] \cdot \dot{U}(t) + [K] \cdot U(t) = F(t)$$

To solve this system numerically, two fundamental tasks must be performed. First, the damping properties of the structure must be approximated, since they are difficult to determine directly from material constants. Second, the solution must be advanced through time using step-by-step numerical integration schemes, whose characteristics strongly influence both the computational cost and the accuracy of the response.

Rayleigh Damping Model

In practical structural applications, the global damping matrix C is exceptionally difficult to determine directly from the material constants. Thus, to circumvent this, Rayleigh Damping is assumed, expressing C as a linear combination of the mass and stiffness matrices:

$$[C] = a[M] + b[K]$$

Where $a[s^{-1}]$ and $b[s]$ are scalar constants. Exploiting the $[M]$ -orthogonality of the eigenvectors $\vec{\Phi}_i$, this relation can be transformed into modal coordinates, yielding the damping ratio ζ_i for any given mode i .

$$\zeta_i = \frac{a}{2\omega_i} + \frac{b\omega_i}{2}$$

In the current project, a constant damping ratio of is prescribed for the first two natural modes. This sets up a linear system of two equations with two unknowns:

$$\begin{bmatrix} 1/2\omega_1 & \omega_1/2 \\ 1/2\omega_2 & \omega_2/2 \end{bmatrix} \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \zeta_1 \\ \zeta_2 \end{bmatrix}$$

The Newmark- β Method (Implicit)

The Newmark method is a implicit family of single step integration approximations that updates displacements and velocities at step $n + 1$ using Taylor series expansions modified by quadrature parameters γ and β :

$$U^{n+1} = U^n + \Delta t \dot{U}^n + \Delta t^2 \left[\left(\frac{1}{2} - \beta \right) \ddot{U}^n + \beta \ddot{U}^{n+1} \right]$$

$$\dot{U}^{n+1} = \dot{U}^n + \Delta t \left[(1 - \gamma) \ddot{U}^n + \gamma \ddot{U}^{n+1} \right]$$

To preserve energy conservation and satisfy second-order accuracy without introducing artificial numerical damping, the average acceleration method (or trapezoidal rule) parameters are chosen:

$$\gamma = \frac{1}{2} \text{ and } \beta = \frac{1}{4}$$

Substituting these expansions into the equation of motion at t^{n+1} yields an effective static algebraic system solved at each step:

$$[K] = [K] + \frac{1}{\beta \Delta t^2} [M] + \frac{\gamma}{\beta \Delta t} [C]$$

$$F^{n+1} = F^{n+1} + [M] \cdot \left(\frac{1}{\beta \Delta t^2} U^n + \frac{1}{\beta \Delta t} \dot{U}^n + \left(\frac{1}{2\beta} - 1 \right) \ddot{U}^n \right) + [C] \cdot \left(\frac{\gamma}{\beta \Delta t} U^n + \left(\frac{\gamma}{\beta} - 1 \right) \dot{U}^n + \frac{\Delta t}{2} \left(\frac{\gamma}{\beta} - 2 \right) \ddot{U}^n \right)$$

Solving the linear system for the next state of displacement:

$$[K] \cdot U^{n+1} = F^{n+1}$$

Stability Note: Under the specific parameter assignment of (the Constant Average Acceleration operator), the Newmark scheme satisfies the algebraic condition for unconditional stability:

$$\gamma \geq \frac{1}{2} \text{ and } \beta \geq \frac{1}{4} \left(\gamma + \frac{1}{2} \right)^2$$

Consequently, the numerical solution is mathematically bounded and guaranteed not to diverge amplificatively, regardless of how large the temporal discretization step Δt is chosen. The selection of the time step is thus completely decoupled from numerical stability boundaries and is governed strictly by the sampling accuracy required to resolve the higher-order harmonic

content of the structural response without introducing significant period elongation or artificial phase errors.

The Central Difference Method

The Central Difference Method is an explicit integration scheme based on replacing the time derivatives of displacement with central finite difference approximations at time t^n :

$$\ddot{U}^n = \frac{1}{\Delta t^2} (U^{n+1} - 2U^n + U^{n-1})$$

$$\dot{U}^{n+1} = \frac{1}{2\Delta t} (U^{n+1} - U^{n-1})$$

Substituting these finite difference operators into the governing equation of motion evaluated strictly at time t^n yields:

$$\left(\frac{1}{\Delta t^2} [M] + \frac{1}{2\Delta t} [C] \right) U^{n+1} = F^n$$

Where F^n corresponds to the residual vector calculated from the following formula:

$$F^n = \vec{F}^n - \left([K] - \frac{2}{\Delta t^2} [M] \right) U^n - \left(\frac{1}{\Delta t^2} [M] - \frac{1}{2\Delta t} [C] \right) U^{n-1}$$

This highlights the core explicit advantage: if a lumped mass matrix is implemented and damping is omitted or diagonalized, the coefficient matrix on the left-hand side becomes strictly diagonal. Computing simplifies to a trivial scalar division per degree of freedom, bypassing the expensive linear-system solution required by implicit methods.

Stability Note: Unlike Newmark- β method, the Central Differential Method is conditionally stable. The numerical solution will wildly diverge (“Blow up”) unless the time step strictly satisfies the Courant-Friedrichs-Lewy (CFL) stability criterion:

$$\Delta t \leq \Delta t_{crit} = \frac{2}{\omega_{max}}$$

Where ω_{max} is the highest natural angular frequency of the assembled mesh system. This demands an exceptional small-time step, especially when using a refined mesh or a consistent mass matrix formulation. Good practice in problems of finite element analysis, where there are many degrees of freedom, is assuming that the first N eigenvalues can approach correctly the transient behavior of the structure.

4.2 Results and Comparisons

Transient analysis results

To validate the accuracy of the custom transient Finite Element Analysis (FEA) solver, a dynamic simulation was performed on the same steel beam, as in the eigenvalue problem. The structure was discretized also with the same meshing method, and divisions along each axis. The dumping coefficients on the Rayleigh method were computed from the first two eigenfrequencies of the custom program as $a = 26.331[1/s]$ and $b = 5.2135 \cdot 10^{-5}[s]$.

The loading application is straight forward both on ANSYS and the custom program. In ANSYS the user has to separate the loading profile into two distinguished parts. The first one corresponds to the harmonic load for the duration of 0.5[s]. Then the second one is the free oscillation of the beam. In the scope of the custom program, the load application was made with a user-defined function F that is evaluated in the defined time step.

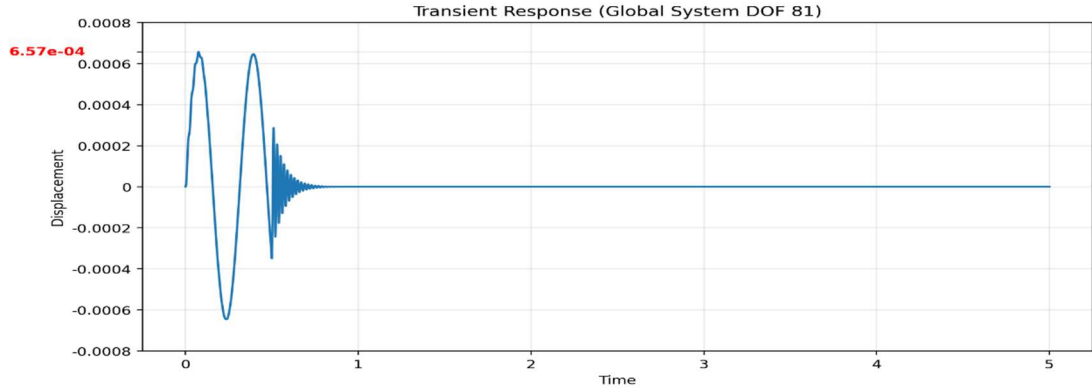


Figure 9. First node oscillation at Y axis (Program)

Regarding the time integration, both the custom solver and ANSYS were implemented using the Newmark- family with a constant time step of $1 \cdot 10^{-4}$ [s]. However, while the custom solver utilizes a classic, undamped Newmark formulation, ANSYS applies a default amplitude decay factor to introduce numerical damping and suppress high-frequency noise. Therefore, observed differences in higher-frequency transient responses are expected due to this numerical dissipation, while broader kinematic deviations can be attributed to implementation details.

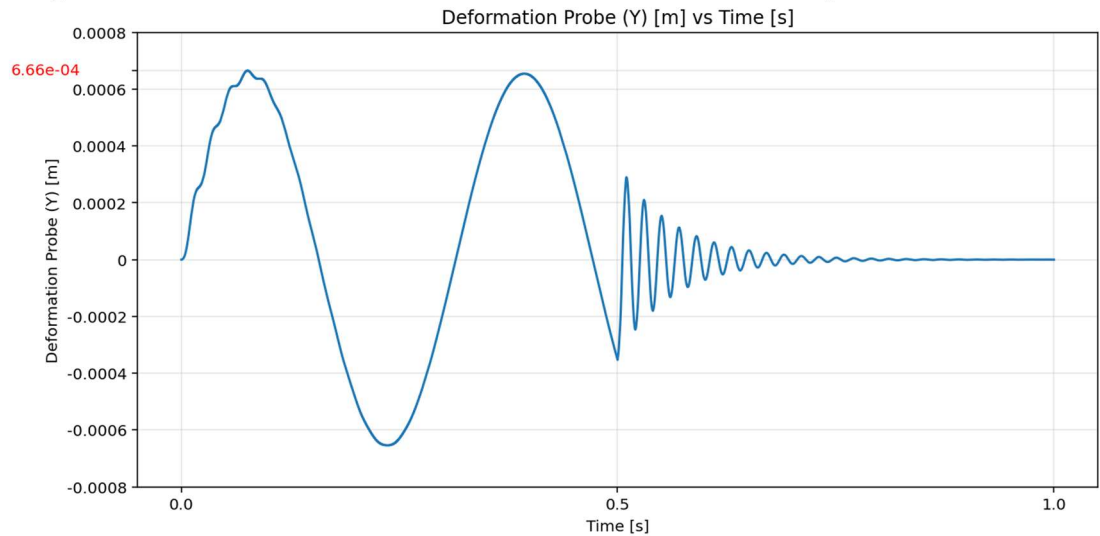


Figure 10. First node oscillation at Y axis (ANSYS)

To compare the transient response between the two programs, two representative nodes were selected along the beam cross-section at the span of the applied load. The first node coincides with the load application point and captures the direct dynamic input together with the local structural response. This signal is expected to contain short-lived high-frequency components superimposed on the global vibration response, which can indeed be observed during the first time steps in both the custom program and the ANSYS results. The second node was selected on the opposite boundary of the beam cross-section. Because it is farther from the point of load application, it reflects the global oscillatory response more clearly and is less influenced by localized numerical effects.

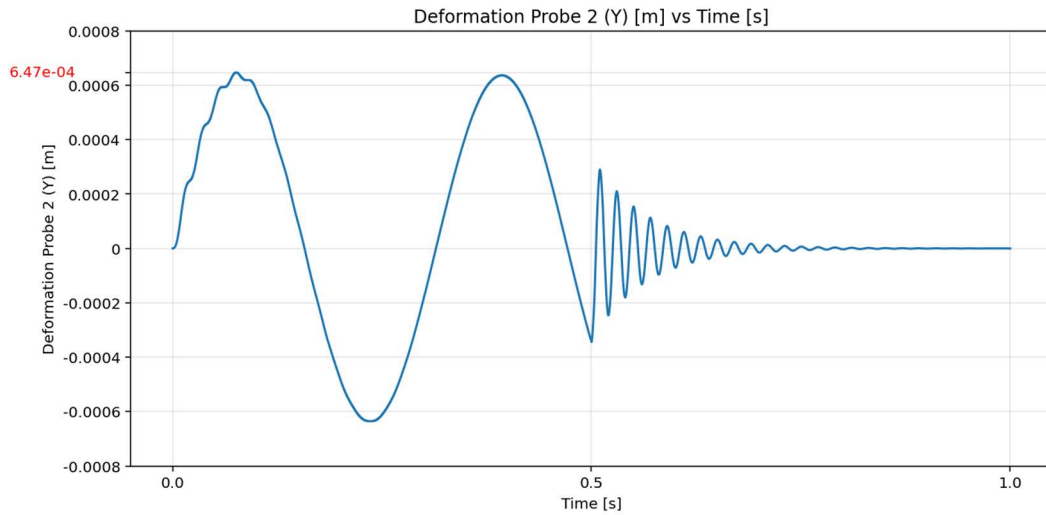


Figure 11. Second node oscillation at Y axis (Program)

A close inspection of the first response curve shows, indeed the presence of short initial high-frequency oscillations at the beginning of the time history. Apart from this localized effect, the custom transient solver reproduces the ANSYS response with high accuracy, confirming the validity of the implementation for the present benchmark case.

It should also be noted that the total simulated time span of 5 [s] is relatively long for the present response, since the oscillations appear to have decayed almost completely after approximately 1 [s]. Therefore, the later part of the time history adds limited additional information regarding the dominant transient behavior, while the essential dynamic features are already captured within the first second of the response.

Comparison between Newmark- β and Central Differences

One of the main distinctions of the two methods is that Newmark method is implicit, while the Central Differences method is explicit. It has already been discussed the effect of that characteristic in the convergence of each algorithm. In addition, the Central differences method seems to struggle with the time step, while the CFL criterion is handy to determine the biggest time-step, in practice the method started to converge to a result after some magnitudes smaller than the limit that was provided. More specifically, in the next table is depicted the biggest time step that resulted in a converged solution for both the Newmark- β and Central Differences methods. In addition, in order to determine the accuracy of the two models it is decided to use the deviation of the ANSYS results about the maximum amplitude of the forcing oscillation at the first 0.5[s].

Table 4. Comparison of different time integrating schemes

Method	Minimum Time-step [s]	Accuracy of the max. Amplitude (Compared to ANSYS)
Newmark- β	$1 \cdot 10^{-4}$	-3.4[
Central Differences	$1 \cdot 10^{-5}$	-5.4[

According to Table 4, the Newmark- β method achieved convergence with a significantly larger time step than the Central Differences method, while also preserving a very high accuracy in the prediction of the maximum response amplitude. This result is consistent with the fundamental distinction between the two schemes: the implicit Newmark- β formulation remains numerically robust for comparatively large time increments, whereas the explicit

Central Differences method is much more sensitive to the temporal discretization. It should be emphasized that the time-step values reported in the table correspond to the largest values for which each method still converged to an acceptable solution. In this sense, the table reflects not only the accuracy of the two schemes, but also their practical stability limits for the present benchmark problem.

At the same time, the CFL criterion presented earlier proved useful in identifying the correct order of magnitude of the critical time step for the Central Differences method. Although the actual converged time step had to be chosen somewhat smaller than the theoretical limit, the criterion still provided a reliable estimate of the stability threshold. A plausible explanation for this behavior is the presence of high-frequency waves generated at the very beginning of the load application. These short-lived oscillatory components are difficult to resolve with a time step chosen too close to the theoretical critical value, and they can therefore force the explicit algorithm to operate with a slightly smaller increment in practice. Consequently, the discrepancy between the theoretical critical time step and the practically convergent one should not be interpreted as a contradiction of the CFL estimate, but rather as an indication of the additional temporal resolution required to capture the initial high-frequency transient response.

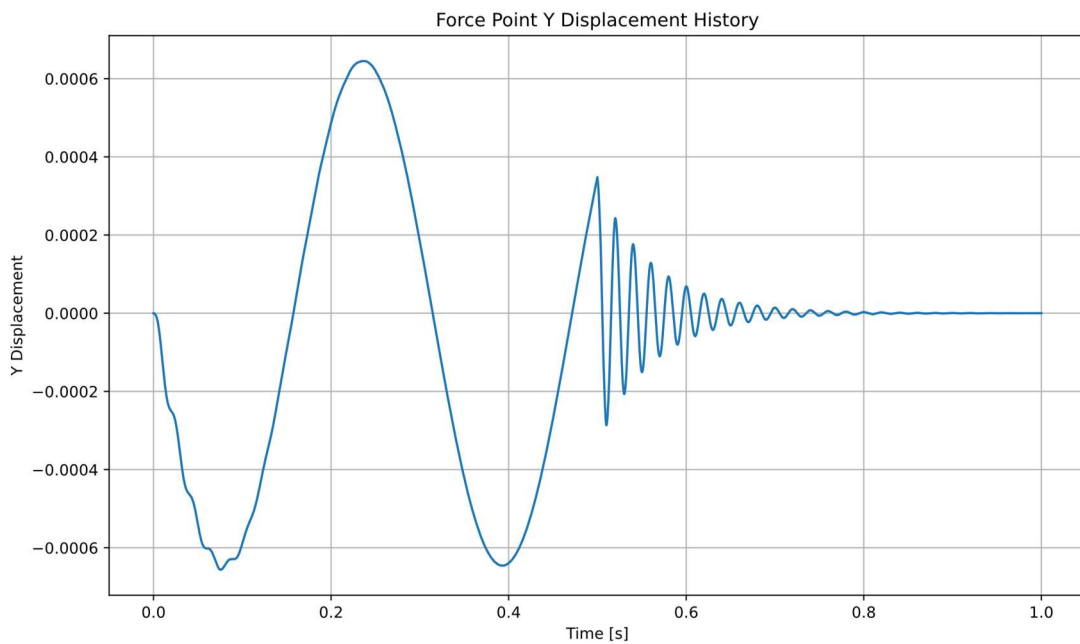


Figure 12. Node 1 oscillation, Central differences (Program)

4.3 Discussion

The influence of Shear locking and Hourglass modes

The previous studies depict a well working algorithm where no artificial stiffening has been influencing the results. However, if the mesh resolution is reduced below a certain limit, the solution starts to depend strongly on the discretization. Beyond this threshold, a significant amount of shear locking is observed within the elements of the mesh. A characteristic example is a coarse discretization, such as a 3 by 12 mesh, where the resulting frequencies begin to show clear inaccuracies caused by shear locking. These locking mainly affect static and transient analyses, where they may lead to nonphysical deformation patterns, artificial stiffening, or instability in the time response. The following diagram shows the resulted oscillation for the new beam where it is observed that the maximum value has been underestimated, since shear locking produces an artificial stiffening. Ultimately, the last diagram, figure 14 was shown that with reduced integration the error became bigger, although still could approach the correct solution form. However the reduced integration method, produce Hourglass mode inside the modes from the eigenvalue problem as shown in the figure 15.

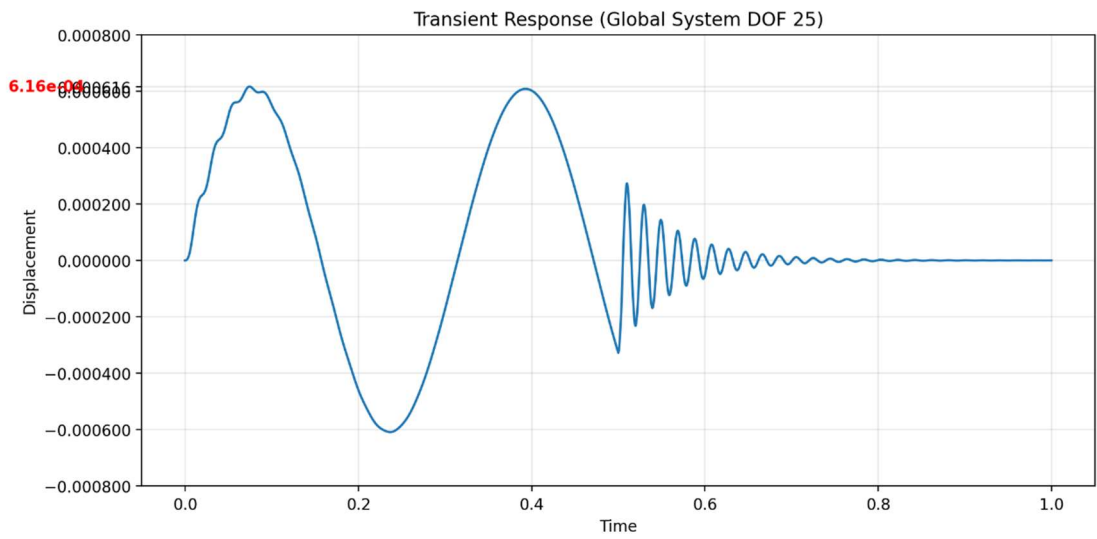


Figure 13. Oscillation of first node, 3x12 Mesh (Program)

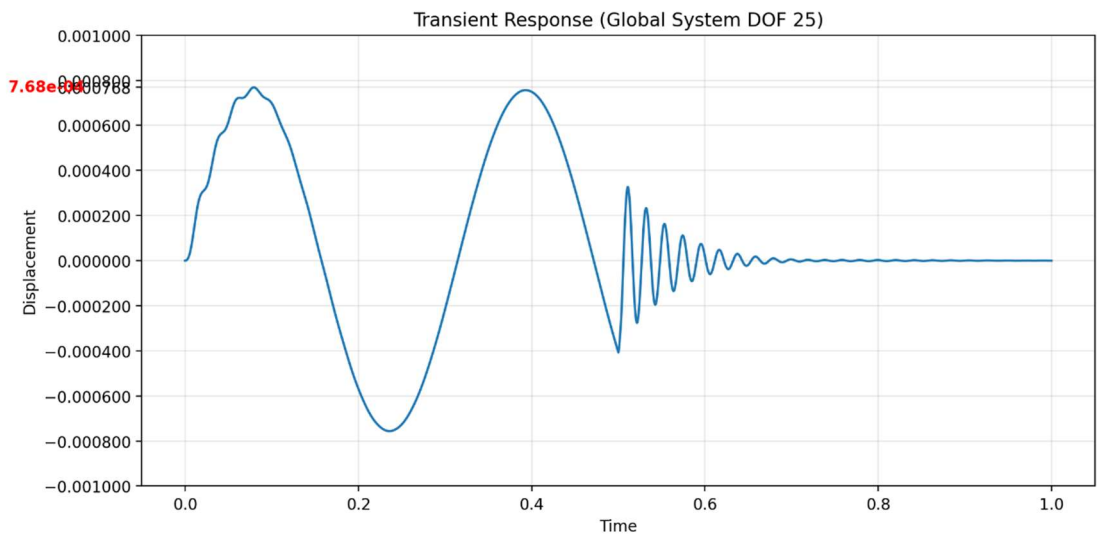


Figure 14. Oscillation of first node, 3x12 Mesh, reduced integration (Program)

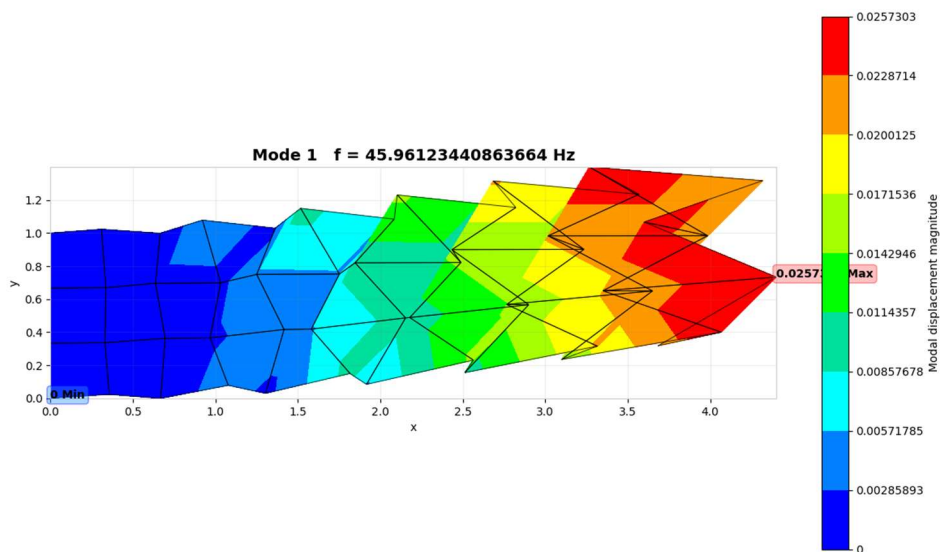


Figure 15. Mode 1, reduced integration method (Program)

Chapter III

Part II

5.1 Continuum Mechanics: Theory of Large Deformations

When a structure undergoes significant displacement or rotations, the fundamental assumptions of infinitesimal strain theory - that the undeformed and deformed configurations are virtually identical - are no longer valid. Consequently, the equilibrium equations must account for the changing geometry, necessitating a mathematically rigorous description of finite kinematics. This is typically formulated using a Total Lagrangian approach, wherein all kinematic and static variables are referred to the initial, undeformed configuration of the body. Let $\vec{X}$ denote the position vector of a material point in the initial configuration and $\vec{x}$ its corresponding position in the deformed configuration. The displacement field is defined as $U = \vec{x} - \vec{X}$. The fundamental measure of deformation is the deformation gradient tensor, F , which maps infinitesimal line elements from the reference to the current configuration, expressed as:

$$[F] = \frac{\partial \vec{x}}{\partial \vec{X}} = I + \nabla_{\vec{X}} U$$

To establish a strain measure that is invariant under rigid body rotation, the right Cauchy - Green deformation tensor $[C] = F^T \cdot F$, is utilized to define the Green - Lagrange strain tensor, $[\varepsilon_{GL}]$. This tensor captures the exact geometric non-linearity arising from the kinematics and is given by the relation:

$$[\varepsilon_{GL}] = \frac{1}{2}([C] - I) = \frac{1}{2} \left(\nabla_{\vec{X}} U + (\nabla_{\vec{X}} U)^T + (\nabla_{\vec{X}} U)^T \cdot \nabla_{\vec{X}} U \right)$$

The presence of the quadratic term $(\nabla_{\vec{X}} U)^T \cdot \nabla_{\vec{X}} U$ introduces a fundamental geometric non-linearity into the strain-displacement relations, coupling the various deformation modes mathematically. In the Total Lagrangian Formulation, the stress measure that is work-conjugate to the Green - Lagrange strain is the Second Piola-Kirchhoff stress tensor, S . The principle of virtual work, representing the weak form of the equilibrium equations over the initial volume V_0 , is expressed as:

$$\int_{V_0} [S] : \delta[\varepsilon_{GL}] dV = \delta W_{ext}$$

Because the strain $[\varepsilon_{GL}]$ is non-linear with respect to the elemental displacement vector U_e , the resulting internal force vector $\vec{F}_{int}(U_e)$, obtained following the finite element spatial discretization, is highly non-linear function of the nodal displacements.

5.2 Numerical Solution Strategy: Load Control

The finite element discretization of the non-linear weak form yields a system of non-linear algebraic equations where the residual force vector $\vec{R}(U)$ must equate to zero. This reaction is defined as:

$$\vec{R}(U) = \vec{F}_{ext} - \vec{F}_{int}(U) = \vec{0}$$

Here, $\vec{F}_{ext}$ is the external force vector and $\vec{F}_{int}(U)$ represents the internal force vector. Because $\vec{F}_{int}(U)$ depends non-linearly on the displacement field U , the system cannot be solved directly in a single step. Instead, an iterative procedure known as load control, or force control, combined with the Newton-Raphson method, is utilized to trace the equilibrium path.

Under the load control strategy, the total external load is applied incrementally, and it is governed by a load multiplier λ_j , such as the external force $\vec{F}_{ext}^{(m)} = \lambda_m \vec{F}_{ref}$, where $\vec{F}_{ref}$ is a reference load vector. Within each specific load increment, the external force is held constant, and the Newton-Raphson method is applied to iteratively minimize the residual $\vec{R}(U)$. For the k-th iteration of a given load step, the non-linear residual equation is linearized using a first-order Taylor series expansion around the current displacement estimate $U^{(k)}$. This linearization is expressed as:

$$\vec{R}(U^{(k)} + \delta U^{(k)}) \simeq \vec{R}(U^{(k)}) + \frac{\partial \vec{R}}{\partial U_{U^{(k)}}} \delta U^{(k)} = \vec{0}$$

The derivative of the residual vector with respect to the displacements yield the tangent stiffness matrix, $[K_T]$. Assuming that the external loading is conservative and independent of the structural deformation, the tangent stiffness matrix is defined strictly by the derivative of the internal forces:

$$[K_T](U^{(k)}) = -\frac{\partial \vec{R}}{\partial U} = \frac{\partial \vec{F}_{int}}{\partial U}$$

The matrix $[K_T]$ is dynamic and encapsulates both the material stiffness and the geometric stiffness, the latter arising from the non-linear strain terms and the initial stress state at that specific iteration. To find appropriate iterative displacement correction, $\delta U^{(k)}$, the linearized system is solved algebraically:

$$[K_T](U^{(k)}) \cdot \delta U^{(k)} = -\vec{R}(U^{(k)})$$

Following this resolution, the total displacement vector for the system is updated according to the relation $U^{(k+1)} = U^{(k)} + \delta U^{(k)}$. This iterative prediction and correction process continues until the norm of the residual vector $\|\vec{R}(U^{(k+1)})\|$, falls below a strictly defined mathematical tolerance threshold. Once this convergence criterion is met, the structure is considered to have reached a state of static equilibrium for that specific load increment, and the algorithm proceeds to the load step.

5.3 Static Analysis: Large Deformations

To thoroughly evaluate the structural behavior under severe loading, a 2D cantilever beam was analyzed in a state of plain stress. The beam features a length of 6[m], a height of 0.5[m] and a thickness of 20[cm] and is modeled with a linear elastic material possessing a Young's modulus of 50[GPa] and Poisson ratio of 0.3[-]. A substantial transverse point load of 20[MN] was applied at its free end. This study contrasts linear small-displacement theory with a geometrically non-linear large deformation formulation to highlight the mathematical reality of the structure.

The Significance of Large Deformation Assumption

The results show that the kinematic assumption controls the predicted response. The geometrically non-linear model gives a tip displacement of about 0.39–0.41 [m], while the linear model gives 13.6–13.9 [m], for the custom program and ANSYS respectively, as shown in the table 5. Such a gap indicates that the undeformed geometry is no longer an adequate reference for equilibrium once the beam experiences large rotations. In this regime, a small-displacement formulation loses validity and severely overestimates flexibility.

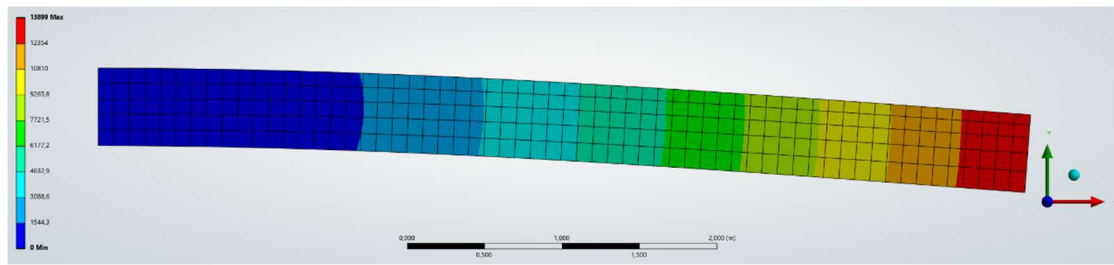


Figure 16. Deformation of beam, Small - Deformation assumption (Program)

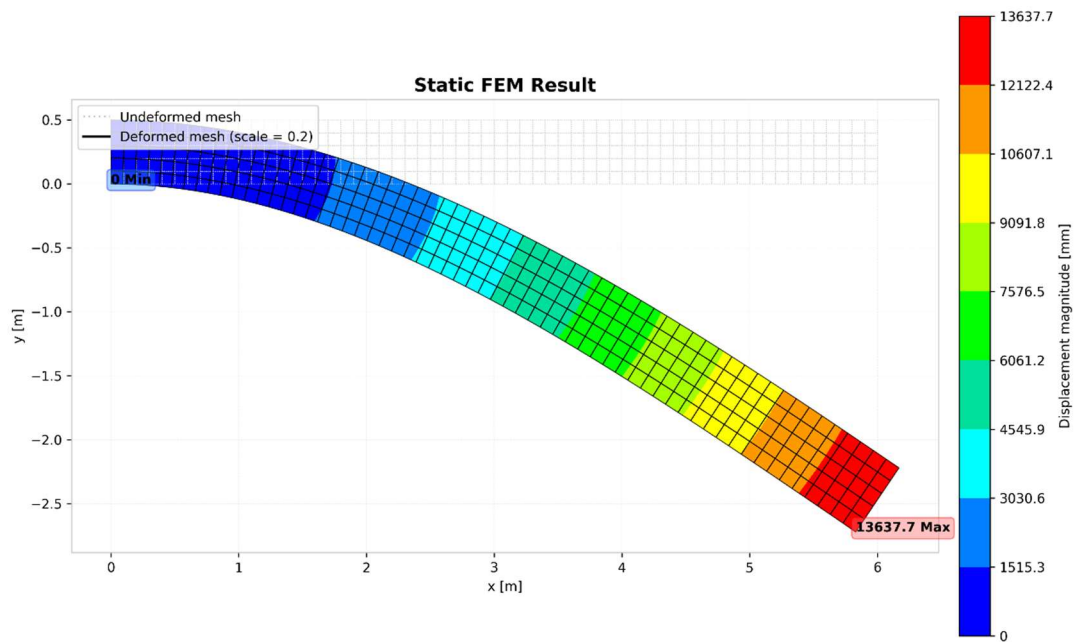


Figure 17. Deformation of the beam, Small- Deformation (Program)

Table 5. Program - ANSYS accuracy comparison, Large Deformations

Approach	Program [m]	ANSYS [m]	Deviation
Large Deformations	4.39	4.41	5.12[%]
Linear	13.61	13.88	1.98[%]

The difference stems from geometric stiffness. In the linear formulation, internal forces are evaluated with respect to the original configuration, so changes in structural geometry do not influence the resisting mechanism. The large-deformation formulation updates equilibrium on the deformed configuration, capturing the additional stiffness produced by beam rotation and axial stretching. As a result, the force–displacement response becomes nonlinear and progressively stiffer, which is consistent with the observed reduction in tip displacement.

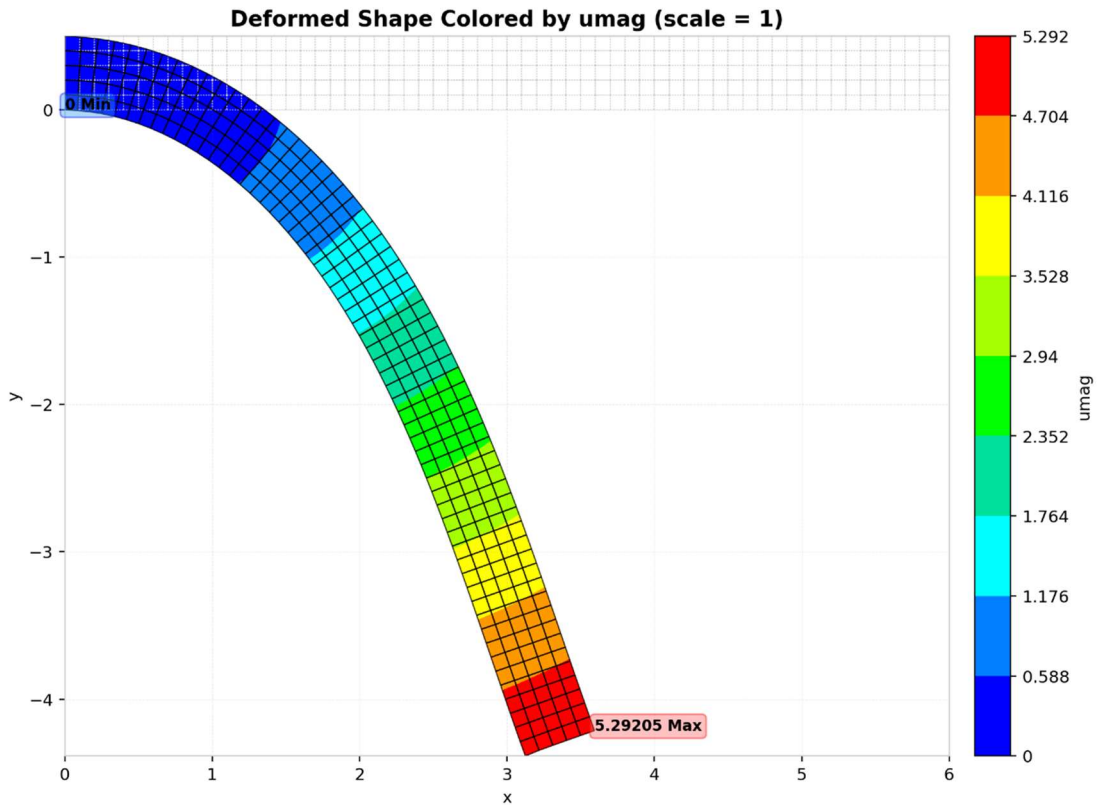


Figure 18. Deformation of beam, Large - Deformation assumption (Program)

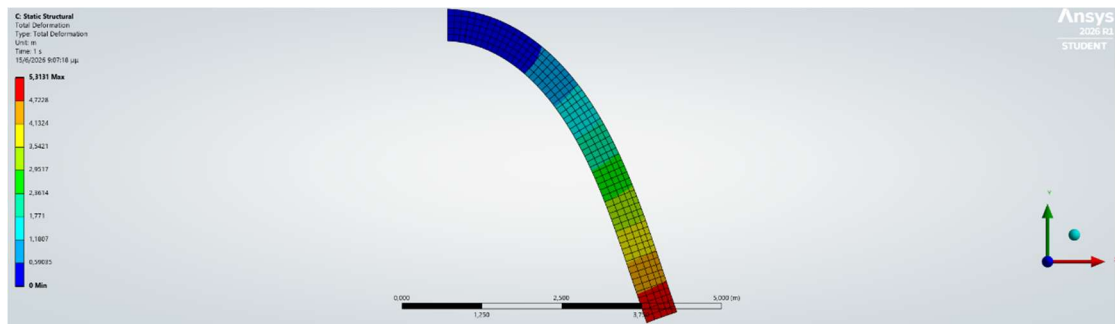


Figure 19. Deformation of beam, Large - Deformation assumption (ANSYS)

Therefore, large-deformation kinematics are required whenever deflections or rotations are no longer small relative to the structural dimensions. In such cases, linear theory may still capture the initial trend, but it cannot provide a reliable final prediction of stiffness or displacement.

The minor residual deviation between ANSYS and the custom program stems from implementation-level nuances rather than fundamental mechanical discrepancies. Although both models utilize identical geometry, material properties, and a comparable discretization of 100 load steps, they diverge in their non-linear solution tracking and convergence criteria. Specifically, the custom program enforces a predefined convergence tolerance of 10^{-8} [%]. In contrast, the ANSYS model utilized default element configurations and active large deformation tracking. Under these program-controlled settings, ANSYS defaults to a load-controlled Newton–Raphson scheme, relying on its internal, dynamically scaled force and displacement L_2 norm tolerances (typically defaulting to 0.5%). Because the software

automatically governs tangent stiffness updates, bisection logic, and the specific equilibrium path followed during the non-linear iterations, slight variations in the force-displacement traversal are entirely expected. Therefore, the observed variance should be interpreted as a standard numerical artifact inherent to comparing two independent finite element architectures, and the strong overall correlation validates the mechanical accuracy of the custom formulation.

Comparison of Reactions

The following table compares the reaction forces at the fixed support of the cantilever beam for both linear and geometrically nonlinear formulations, as obtained from the developed program and ANSYS. Support reactions serve as a global verification metric since they directly reflect overall equilibrium rather than local field variables. While the linear formulation is governed by the undeformed configuration, the geometrically non-linear formulation accounts for changes in geometry during deformation, which may lead to small redistributions of internal forces, particularly under large rotations.

Table 6. Program - ANSYS Reactions comparison

Approach	Program Rx	ANSYS Rx	Program Ry	ANSYS Ry
Large Deformations at last step	$2.2 \cdot 10^{-4}[\text{N}]$	576.13[N]	20[MN]	20[MN]
Linear	$1.1 \cdot 10^{-4}[\text{N}]$	$-8.7 \cdot 10^{-5}[\text{N}]$	20[MN]	20[MN]

As shown in Table 6, excellent agreement is observed between the two implementations. In both formulations, the total vertical reaction force is recovered as Rx and Ry, matching the ANSYS results exactly. The horizontal reaction is negligible in all cases, consistent with the absence of horizontal loading. The small discrepancy in the nonlinear analysis remains several orders of magnitude smaller than the applied load and is therefore not physically significant. This fact is confirmed also by Figure 20, where in contrast with the reaction force at the Y axis, this diagram doesn't show any similar trend. If this part of the force was a result of the loading it would be expected to have the same monotonous incremental behavior. Overall, the results confirm that the proposed implementation correctly enforces global equilibrium in both linear and geometrically nonlinear regimes.

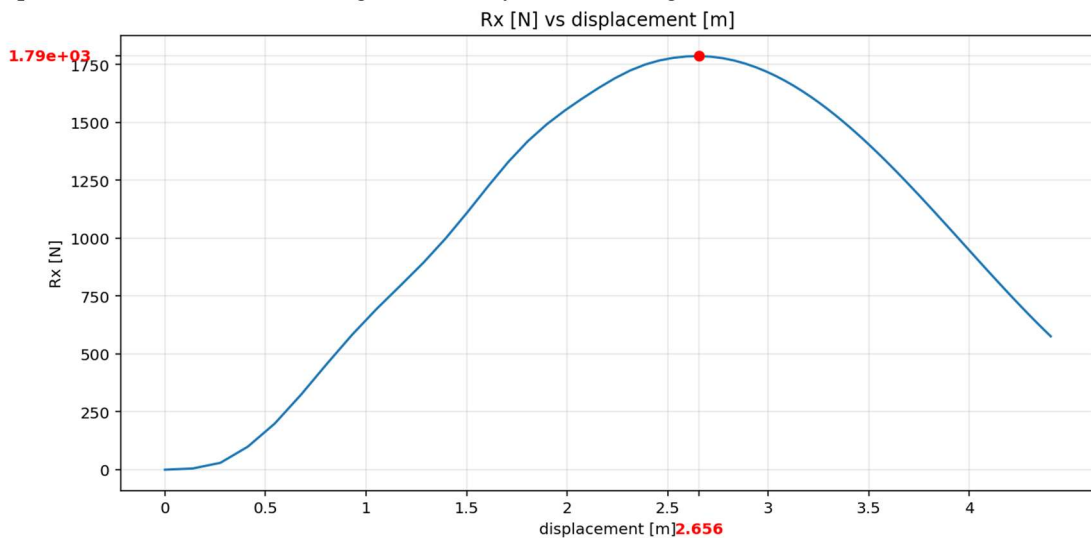


Figure 20. Reaction at X axis during the load- displacement method (ANSYS)

Comparison of Diagrams Force - Deformation

In the following illustration, the applied force diagram is plotting against the displacement of the load application node. As can be clearly observed, the non-linear curve traces an initial slope identical to the linear one. However, as large deformations method gradually bends upward, exhibiting a load-stiffening response. Thus, the linear approach yields after a deformation of the beam that exceeds about $1.5[m]$ or a loading over $2.6[MN]$, which is substantial compared to its size. In this particular point in the specific diagram the error between the two approaches is around 2%. Additionally, the large overestimation of the final displacement at the free end of the beam is depicted once more.

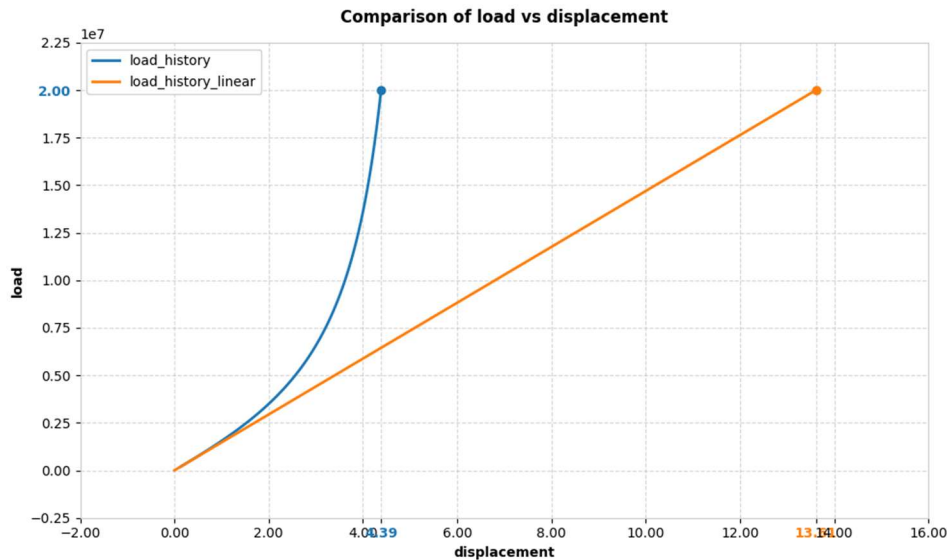


Figure 21. Force - Displacement diagram (Program)

In this case study the load was applied in 1[%] increments, ensuring a stable but quick convergence of the program. While the program was running, the Newton-Raphson iterative algorithm used about 8~9 iterations to achieve a tolerance of $1 \cdot 10^{-8}[\%]$. It was observed that if the increments were smaller than the limit above (~ 10 steps), the load control method converged to the correct result, but the Newton-Raphson method took more iterations for the same accuracy. As a result, the method was quick, but it was not guaranteed to work correctly if a small change was induced in the problem.

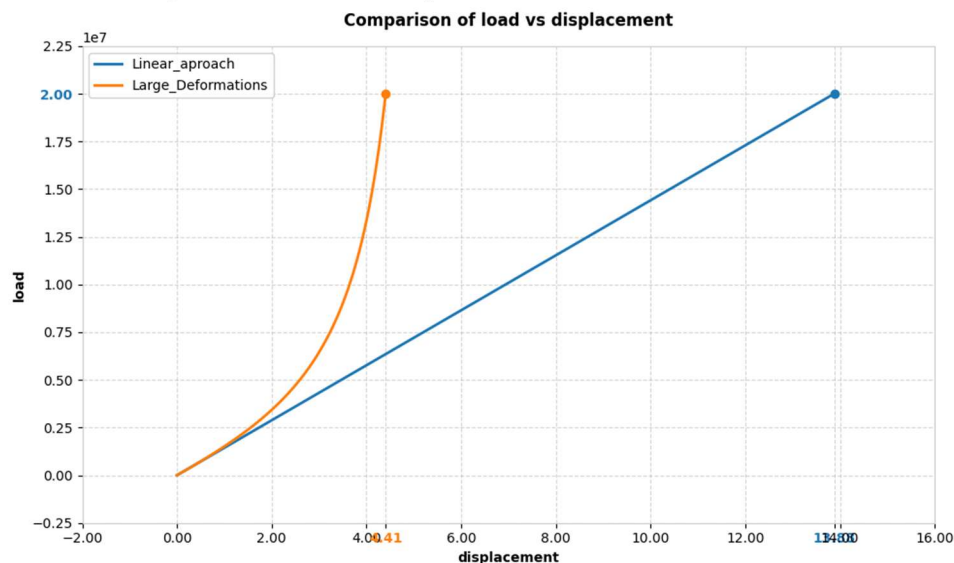


Figure 22. Force - Displacement diagram (ANSYS)

Ultimately the force - displacement graph can also be an evaluation of the correctness of the custom algorithm. Figure 22 shows the corresponding result from the ANSYS tool, where the results for both curves show a minuscule deviation, as shown at the corresponding figure 23.

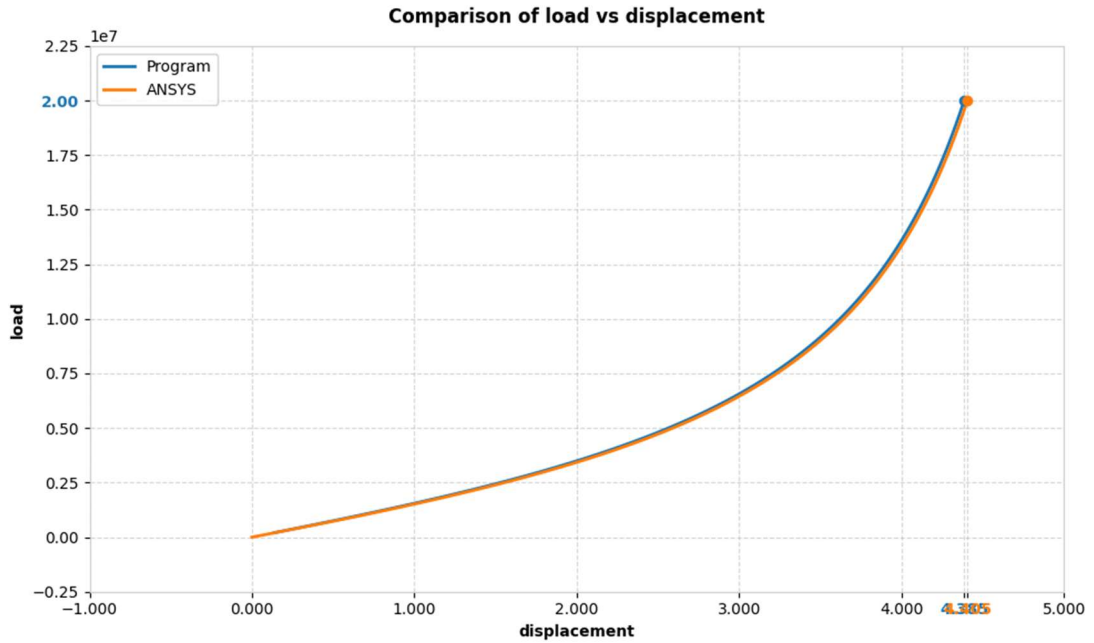


Figure 23. Load control comparison between Program and ANSYS

5.4 Discussion

As observed in the force–displacement curves, the linear and geometrically non-linear solutions coincide closely only during the initial loading stage. In this region, the transverse deflection and rotation of the cantilever remain small, so the undeformed configuration is still an acceptable reference for equilibrium. Consequently, the tangent response is nearly identical in both formulations, and the structural behavior appears approximately linear. This explains why the early portion of the non-linear curve follows the same initial slope as the linear solution.

As the load increases, this agreement is progressively lost because the beam no longer deforms within the assumptions of infinitesimal kinematics. Once the tip displacement becomes non-negligible relative to the beam depth, the structure develops large rotations and additional axial stretching, which introduce geometric stiffening into the response. At that point, the linear model continues to extrapolate the initial flexibility and therefore overestimates the final displacement, whereas the large-deformation formulation updates the stiffness and equilibrium path consistently with the evolving geometry. The numerical comparison with ANSYS confirms this interpretation: both models produce almost identical non-linear trends and similar support reactions, while the small remaining discrepancies can be explained by differences in element implementation, numerical integration, and solver settings. Overall, the study shows

that large-deformation kinematics are not simply a refinement of the linear model, but a necessary requirement for obtaining a physically meaningful prediction in strongly bent cantilever problems.

Chapter IV

Part III

6.1 Continuum Mechanics: Theory of Neo-Hookean Hyperelastic Materials

For hyperelastic materials, the constitutive response is derived from a strain-energy density function, which defines the stored elastic energy per unit reference volume. In this class of materials, the stress state is obtained directly from the energy potential, providing a thermodynamically consistent constitutive framework for finite deformations:

$$W([F](X), X) = \int_{t_0}^t [P]:[\dot{F}]dt, \quad \dot{W} = [P]:[\dot{F}]$$

Equivalently, this constitutive relation can be expressed in differential form, where the variation of the strain-energy density is related to the work-conjugate stress and strain measures:

$$\dot{W} = \frac{\partial W}{\partial [F]}: [\dot{F}] \Rightarrow [P] = \frac{\partial W}{\partial [F]}$$

In hyperelasticity, the internal work is assumed to be stored entirely as recoverable elastic energy. Since the material behavior is conservative, no energy is dissipated through plastic flow, viscosity, or thermal effects within the constitutive law. As a result, the strain-energy function depends only on the current deformation state and is independent of the deformation path. In the present total Lagrangian formulation, the energy is expressed in terms of the Green–Lagrange strain tensor, which leads naturally to the Second Piola–Kirchhoff stress as the work-conjugate stress measure.

$$W([F](X), X) = W([C](X), X)$$

Accordingly, when the strain-energy density function is written as, differentiation with respect to the Green–Lagrange strain tensor yields the constitutive stress relation:

$$\dot{W} = \frac{\partial W}{\partial [C]}: [\dot{C}] = \frac{1}{2} [S]: \dot{C} = [S]: [\dot{\varepsilon}_{GL}]$$

Or equally written as:

$$[S] = 2 \frac{\partial W}{\partial [C]} = \frac{\partial W}{\partial [\varepsilon_{GL}]}$$

For Neo-Hookean materials, the strain-energy density function takes the following form:

$$W = \frac{\mu}{2} (J^{-2/3} I_1 - 3) + \frac{k}{2} (J - 1)^2$$

Where $I_1 = \text{trace}([C])$ and $J = \det([C])$.

By differentiating this expression with respect to the Green–Lagrange strain tensor, the Second Piola–Kirchhoff stress tensor is obtained as:

$$[S] = \mu J^{-2/3} \left(I - \frac{1}{3} I_1 [C]^{-T} \right) + k(J - 1) J C^{-T}$$

While:

$$\begin{aligned} \frac{\partial I_1}{\partial [C]} &= I \\ \frac{\partial J}{\partial [C]} &= \frac{1}{2} J [C]^{-T} \end{aligned}$$

Unlike linear elasticity, where the constitutive matrix is constant, hyperelasticity leads to a deformation-dependent constitutive response. For this reason, the material tangent tensor is not constant but varies with the current strain state. Its linearization defines the tangent constitutive matrix required in the Newton–Raphson solution procedure:

$$[D] = \frac{\partial [S]}{\partial [\varepsilon_{GL}]} = 2 \frac{\partial [S]}{\partial [C]}$$

Consequently, by carrying out the secondary differentiation of the stress equation, the following fourth-order material tangent constitutive tensor (which maps directly to the code terms from your loop) is obtained:

$$\begin{aligned} \mathbb{C}_{ijkl} = 2 \left[\frac{2}{3} \mu J^{-2/3} I_1 \mathcal{J}_{ijkl}^{\text{inv}} - \frac{2}{3} \mu J^{-2/3} (\delta_{ij} C_{kl}^{-1} + C_{ij}^{-1} \delta_{kl}) \right. \\ \left. + \left(\frac{2}{9} \mu J^{-2/3} I_1 + KJ(2J - 1) \right) C_{ij}^{-1} C_{kl}^{-1} - 2KJ(J - 1) \mathcal{J}_{ijkl}^{\text{inv}} \right] \end{aligned}$$

Where $\mathcal{J}_{ijkl}^{\text{inv}} = \frac{1}{2} (C_{ik}^{-1} C_{jl}^{-1} + C_{il}^{-1} C_{jk}^{-1})$ represents the symmetric fourth-order inverse identity tensor.

To convert the fourth-order material tangent tensor $\mathbb{C}_{ijkl}$ into a 3×3 matrix format (commonly called the material tangent matrix $\mathbf{D}$ in Voigt notation for plane strain conditions), we exploit the minor symmetries of the tensor:

$$\mathbb{C}_{ijkl} = \mathbb{C}_{jikl} = \mathbb{C}_{ijlk}$$

For plane strain problems focusing on the $x - y$ plane, the relevant independent index pairs are 11, 22, and 12. We map these 4D tensor index pairs (i, j) and (k, l) into single matrix row (a) and column (b) indices according to the following convention:

$$\mathbf{D}_{ab} = w_a w_b \mathbb{C}_{ijkl}$$

Where the weight factor w_m (for $m \in \{a, b\}$) scales the shear components by a factor of 2:

$$w_m = \begin{cases} 1 & \text{if } i = j \quad (a, b \in \{1, 2\}) \\ 2 & \text{if } i \neq j \quad (a, b = 3) \end{cases}$$

Applying this mapping yields the explicit structure of the $3 \times 3D$ matrix:

$$\mathbf{D} = \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ D_{21} & D_{22} & D_{23} \\ D_{31} & D_{32} & D_{33} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & 2\mathbb{C}_{1112} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & 2\mathbb{C}_{2212} \\ 2\mathbb{C}_{1211} & 2\mathbb{C}_{1222} & 4\mathbb{C}_{1212} \end{bmatrix}$$

6.3 Prescribed displacement method

Displacement-Controlled Solution Procedure

For problems involving large geometric nonlinearities, the structural response is obtained using a displacement-controlled incremental–iterative procedure. Instead of applying external nodal forces, the loading is introduced through prescribed displacements imposed on selected degrees of freedom. The corresponding reaction forces are recovered from the equilibrium solution after convergence.

The global displacement vector is partitioned into free and prescribed degrees of freedom as

$$U = [U_f, U_p]^T$$

where U_f contains the unknown displacement components and U_p contains the prescribed displacement components.

The target prescribed displacement $\bar{U}_p$ is applied incrementally over N load steps according to

$$U_p^{(n)} = a_n \cdot \bar{U}_p$$

, where:

$$a_n = n/N, n = 1, 2, \dots, N$$

At each step, the prescribed degrees of freedom are fixed at their current values, while the remaining displacement components are determined from the nonlinear equilibrium equations. Since no external nodal force vector is applied, equilibrium is enforced only on the free degrees of freedom. Therefore,

$$\overrightarrow{F_{int,f}}(U) = \vec{0}$$

where $\overrightarrow{F_{int,f}}$ denotes the internal force vector associated with the unconstrained degrees of freedom.

The residual vector is defined as:

$$R_f = -\overrightarrow{F_{int,f}}$$

The nonlinear system is solved using the Newton–Raphson method. Linearization of the equilibrium equations at iteration i yields

$$K_{T,ff}^{(i)} \Delta U_f^{(i)} = R_f^{(i)}$$

, where:

$K_{T,ff}^{(i)}$ = reduced tangent stiffness matrix

$\Delta U_f^{(i)}$ = displacement correction vector

$R_f^{(i)}$ = residual vector

The displacement field is updated according to

$$U_f^{(i+1)} = U_f^{(i)} + \Delta U_f^{(i)}$$

Following each iteration, the prescribed displacements are reimposed exactly,

$$U_p^{(i+1)} = \alpha_n U_p$$

The iterative process continues until the convergence criterion $\|R_f\|/\max(\|F_{int}\|, 1) < \varepsilon$ is satisfied, where ε denotes the prescribed tolerance.

After convergence of a load step, the reaction forces are recovered from the internal force vector. Since no external nodal forces are present, $R_{react} = F_{int}$ for the constrained degrees of freedom.

The displacement-controlled strategy enables stable tracing of the nonlinear equilibrium path and is particularly suitable for analyses involving large deformations, geometric instabilities, and post-critical structural behavior.

6.4 Static Analysis: Comparison Between Linear Elastic and Neo-Hookean Material

During the setup of the two programs, aluminum was assigned to the beam so that the same baseline elastic constants could be used in both the linear elastic and Neo-Hookean formulations. The geometry and mesh were kept consistent with the previous case study, with the difference that a plane strain assumption was adopted here. In plane strain, the out-of-plane strain is constrained to zero, so the 2D model represents a body of uniform depth normal to the analyzed section. Consequently, the computed reaction forces must be interpreted consistently with the assumed unit out-of-plane thickness used by the formulation. For loading, a prescribed displacement of 1.5 [m] was imposed at the rightmost edge.

In this case study, the results obtained from ANSYS, and the custom program are compared in order to assess the implementation of the Neo-Hookean constitutive model and to benchmark its numerical response against a linear elastic model under the same displacement-controlled loading. This comparison should be interpreted primarily as a constitutive-model benchmark rather than as a physically realistic material model for aluminum, since Neo-Hookean hyperelasticity is generally associated with rubber-like materials and not with metallic materials that exhibit elastoplastic behavior. In this sense, the study is intended to highlight the effect of adopting a hyperelastic constitutive law on the global structural response.

For the Neo-Hookean material case, the nonlinear solution was carried out using a convergence tolerance of 0.01[%] over 80 displacement steps. During the initial stages of the solution procedure, the model showed noticeable difficulty in convergence, indicating that the early increments were more sensitive to the nonlinear material response and to the stabilization of the Newton–Raphson iterations.

To initialize a compressible Neo-Hookean formulation from standard linear elastic constants, the material parameters may be related through the small-strain limit of isotropic elasticity. In particular, the initial shear modulus μ and the bulk modulus K are obtained from Young's modulus E and Poisson's ratio ν through the classical relations $\mu = E/2(1 + \nu)$ and $K = E/3(1 - 2\nu)$. These relations ensure that the Neo-Hookean model reproduces the same initial tangent stiffness as the linear elastic model in the small-strain regime; however, they do not imply that the finite-strain response of the Neo-Hookean material is physically representative of aluminum. In ANSYS, the Neo-Hookean model is commonly defined through the initial shear modulus together with a material incompressibility parameter D , for which the corresponding initial bulk modulus satisfies $D = 2/K$. In the following table the corresponding parameters for Aluminum are shown.

Table 7. Neo - Hookean Parameters for Aluminum

Material	Young's Modulus [Pa]	Poisson's Ratio [-]	Initial Shear Modulus [Pa]	Bulk Modulus [Pa]
Aluminum	$50 \cdot 10^9$	0.3	$1.92 \cdot 10^{10}$	$4.17 \cdot 10^{10}$

Displacement Field Comparison

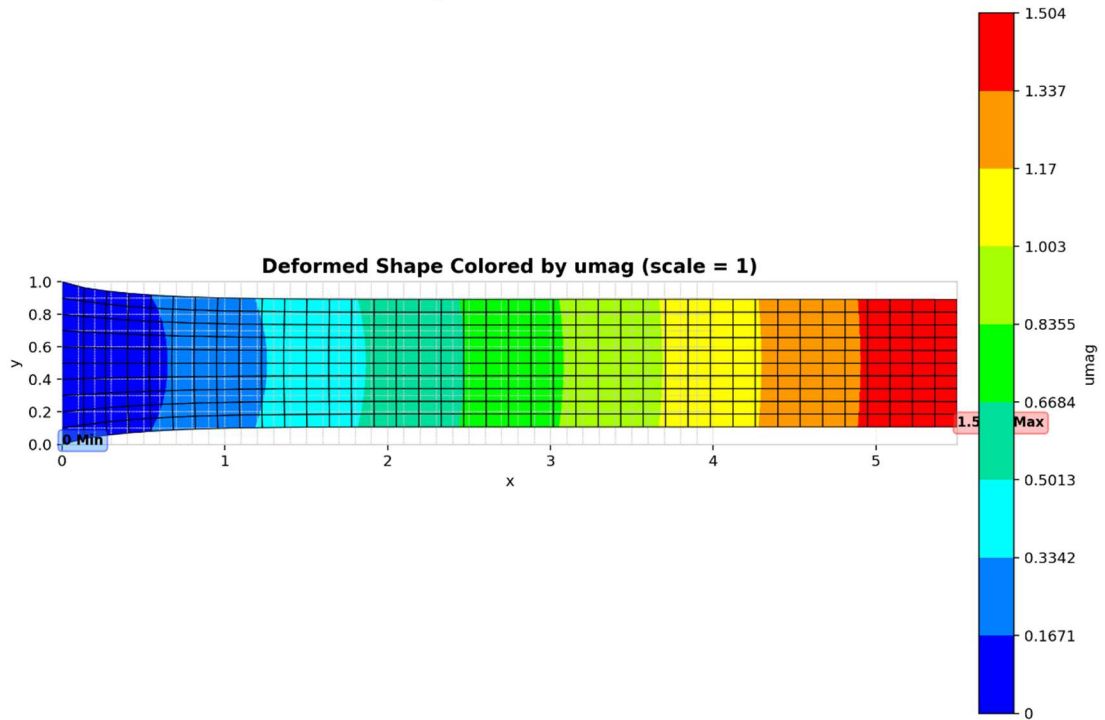


Figure 24. Total deformation of the beam, Neo-Hookean Material (Program)

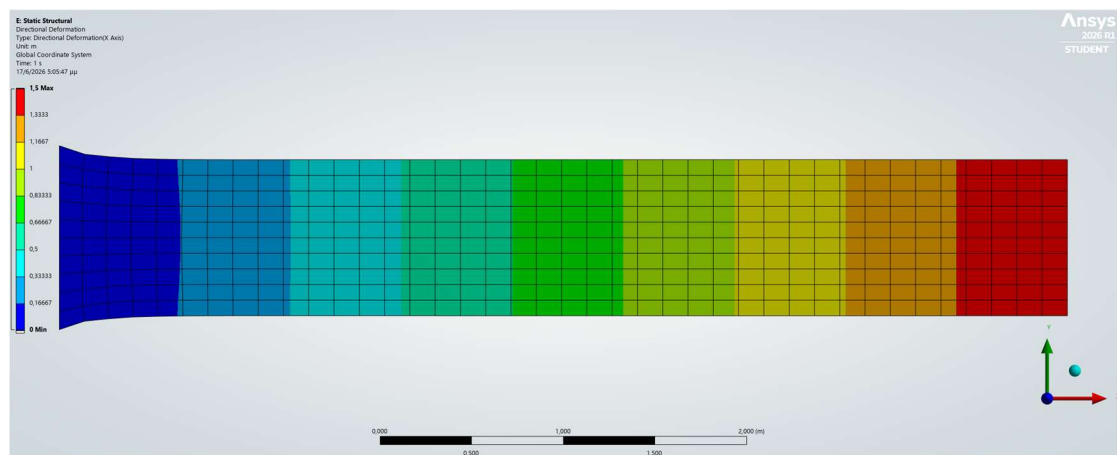


Figure 25. Total deformation of the beam, Neo-Hookean Material (ANSYS)

Figures 24 and 25 show that the total displacement fields obtained for the Neo-Hookean material by the custom program and ANSYS are in very good overall agreement. In both cases, the global deformation pattern is essentially the same, confirming that the implemented constitutive formulation reproduces the expected nonlinear kinematic response under the imposed displacement. Nevertheless, a closer inspection reveals slight local differences between the two solutions, which are concentrated mainly near the fixed support side of the beam. In that region, where the displacement gradients are highest and the constraint conditions strongly influence the local response, the ANSYS solution appears slightly smoother, whereas

the custom program predicts a marginally different distribution of deformation. These discrepancies remain limited and do not alter the overall deformation mode, indicating that the custom implementation captures the Neo-Hookean response with satisfactory accuracy while retaining minor local deviations close to the clamped boundary.

A comparison between the linear elastic, at figures 26 and 27, and Neo-Hookean results further shows that the overall displacement fields do not differ significantly, neither in the ANSYS simulations nor in the custom program. In both implementations, the deformed configuration retains essentially the same global shape, which indicates that for the present boundary conditions and geometry the adoption of the Neo-Hookean constitutive law does not lead to major kinematic deviations with respect to the linear material model. The observed differences remain minor and are mainly local, while the general deformation pattern is preserved in all four cases.

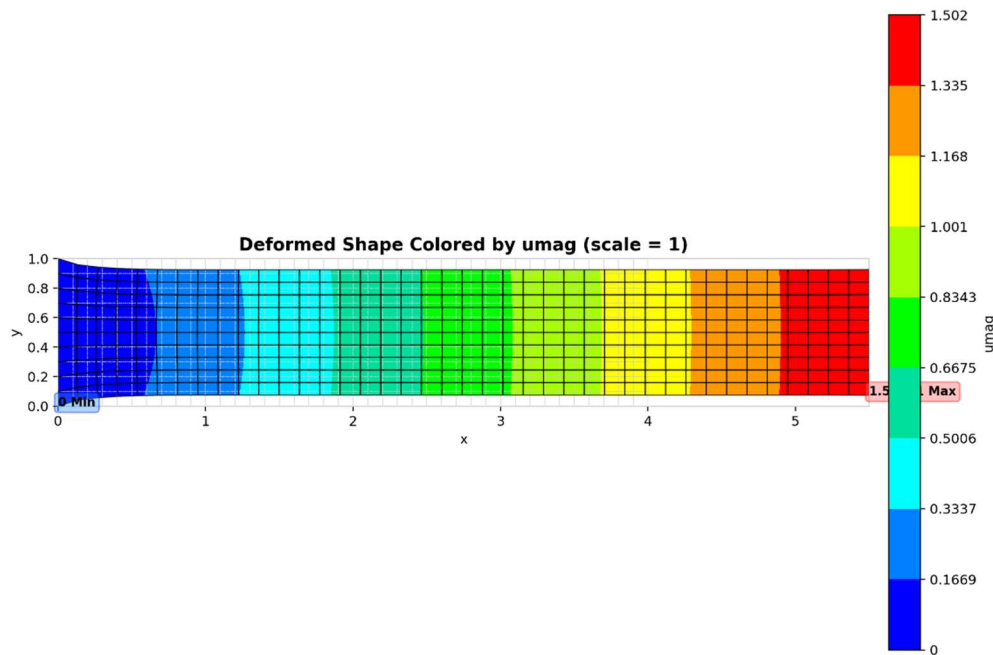


Figure 26. Total deformation of the beam, Linear Material (Program)

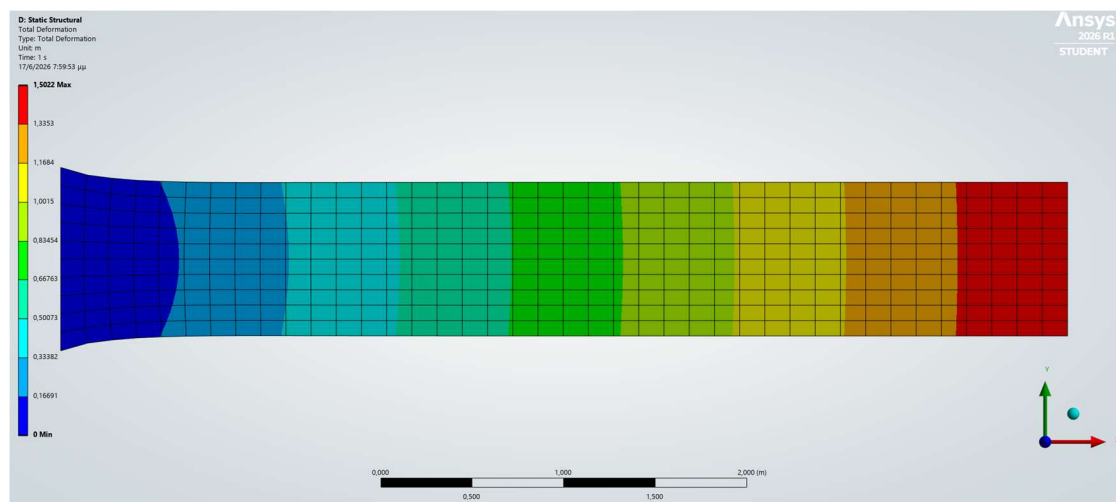


Figure 27. Total deformation of the beam, Linear Material (ANSYS)

The main difference between the two deformation fields is observed in the displacement distribution along the Y axis, as shown in the figures 28 to 31. In particular, near the fixed support, the Neo-Hookean material exhibits slightly larger Y-direction displacements than the corresponding linear elastic material. Away from the clamped boundary, however, the two fields remain very similar, indicating that the influence of the constitutive law is mainly

localized in the support region, where the deformation gradients and constraint effects are more pronounced.

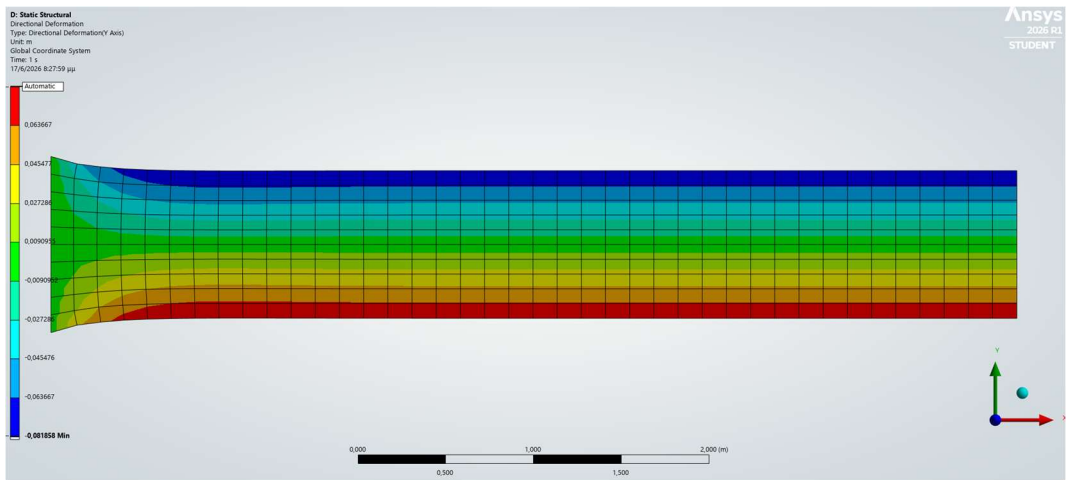


Figure 28. Deformation on Y axis, Linear Material (ANSYS)

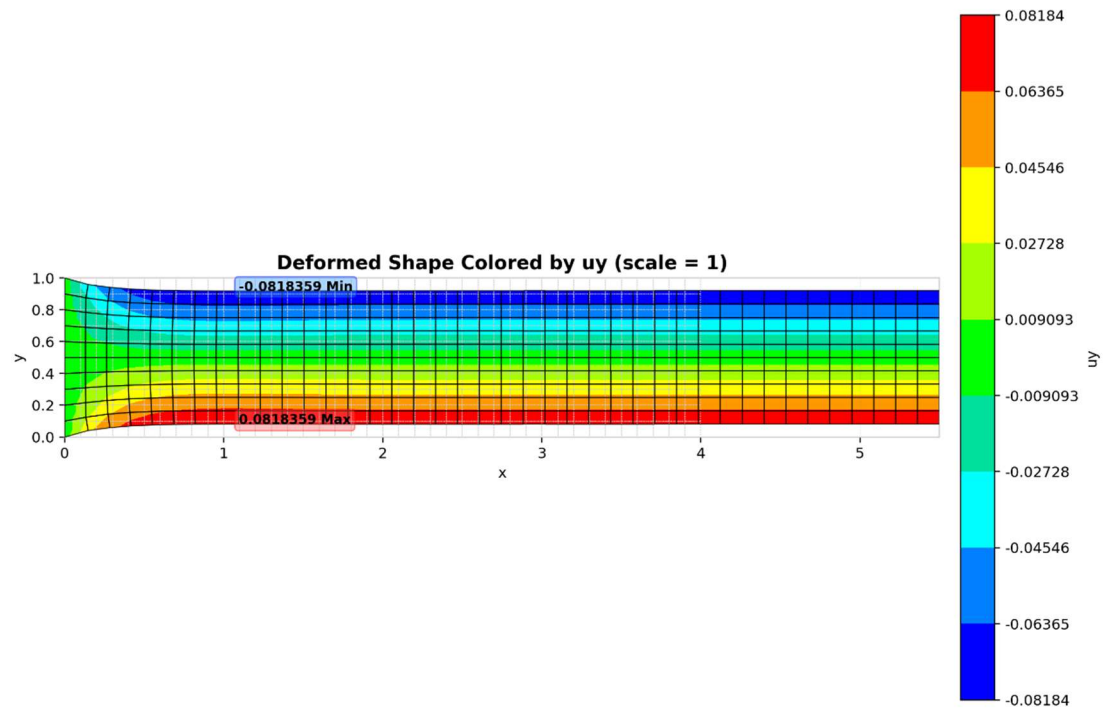


Figure 29. Displacement on Y axis, Linear Material (Program)

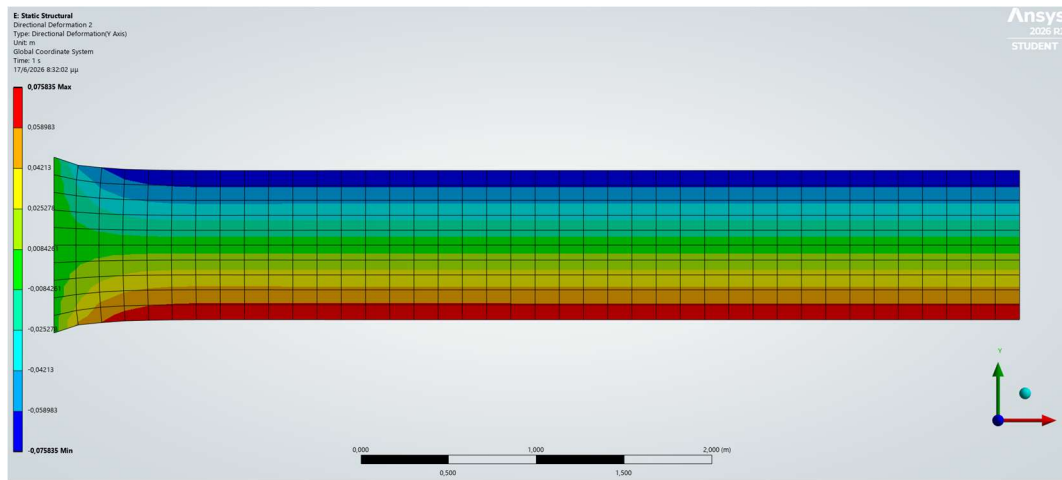


Figure 30. Displacement on Y axis, Neo-Hookean Material (ANSYS)

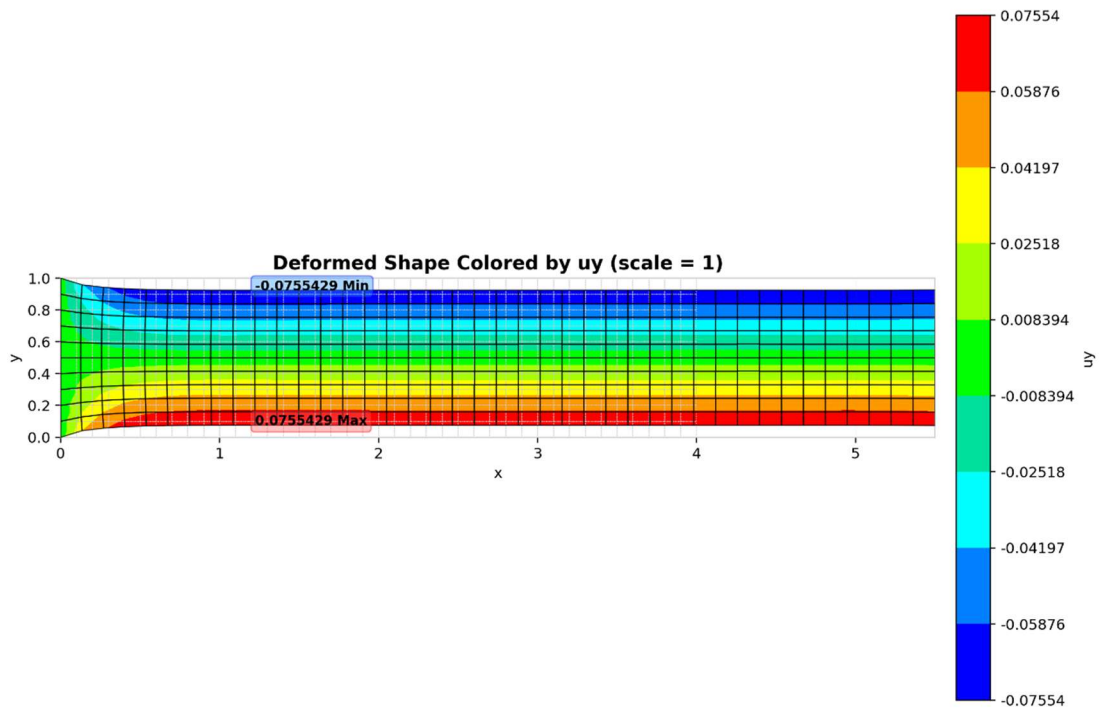


Figure 31. Displacement on Y axis, Neo - Hookean Material (Program)

Stress Field Comparison

Visualizing the stress state in a comprehensive manner is inherently challenging, since the stress field is by nature a tensor field — at every point in a body, the full stress state is described by a second-order tensor with up to six independent components. To address this, engineers have developed failure and distortion criteria that map the complete 2D or 3D stress tensor to a single, physically meaningful scalar quantity known as an equivalent stress. A prominent example is the von Mises equivalent stress, which is rooted in the distortion energy theory and is particularly well-suited for ductile materials, where yielding initiates when the distortion energy density reaches a critical threshold. This scalar representation allows the stress field throughout the entire mesh to be depicted in an intuitive and visually comprehensive manner. Beyond the von Mises criterion, commercial packages such as ANSYS also offer principal stress plots as an alternative equivalent stress representation. As demonstrated in Figures 32 to 35, the custom-developed FEA solver reproduces the stress field obtained from ANSYS with remarkable precision, validating the accuracy of the implemented formulation, once more.

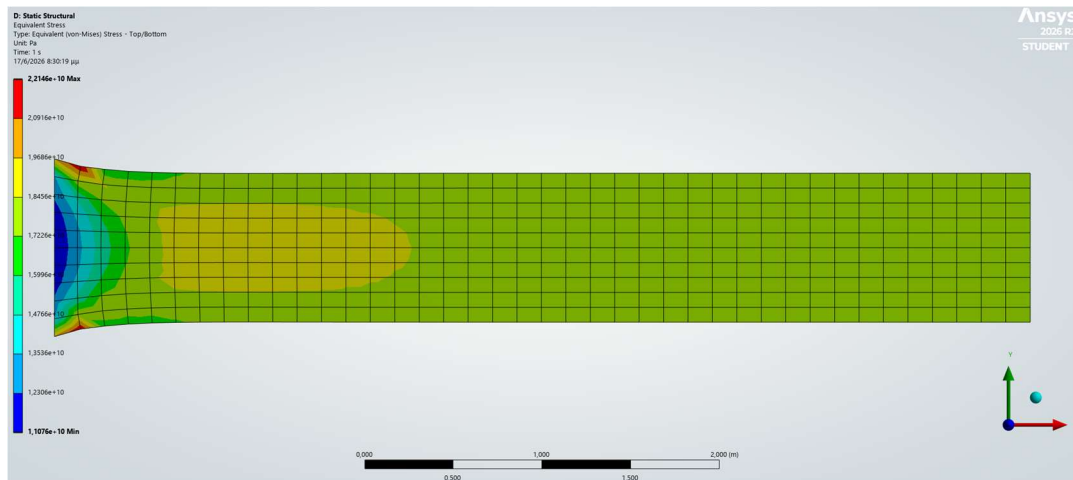


Figure 32. Equivalent von Mises Stress Field, Linear Material (ANSYS)

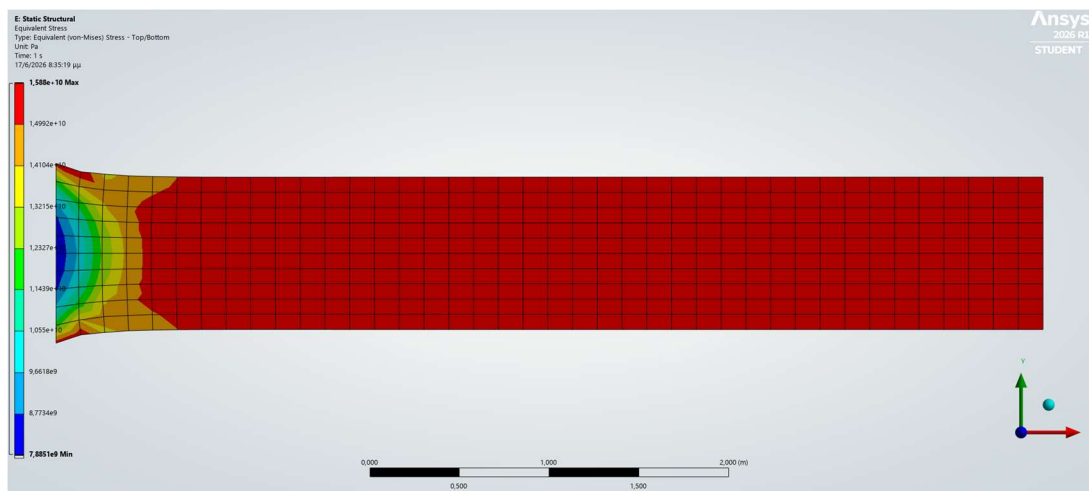


Figure 33. Equivalent von Mises Stress Field, Neo-Hookean Material (ANSYS)

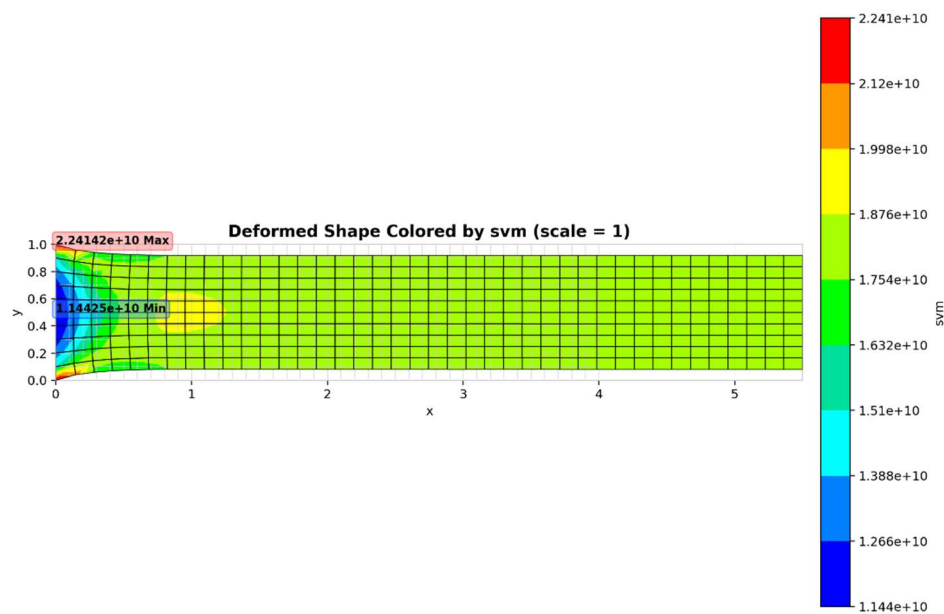


Figure 34. Equivalent von Mises Stress Field, Linear Material (Program)

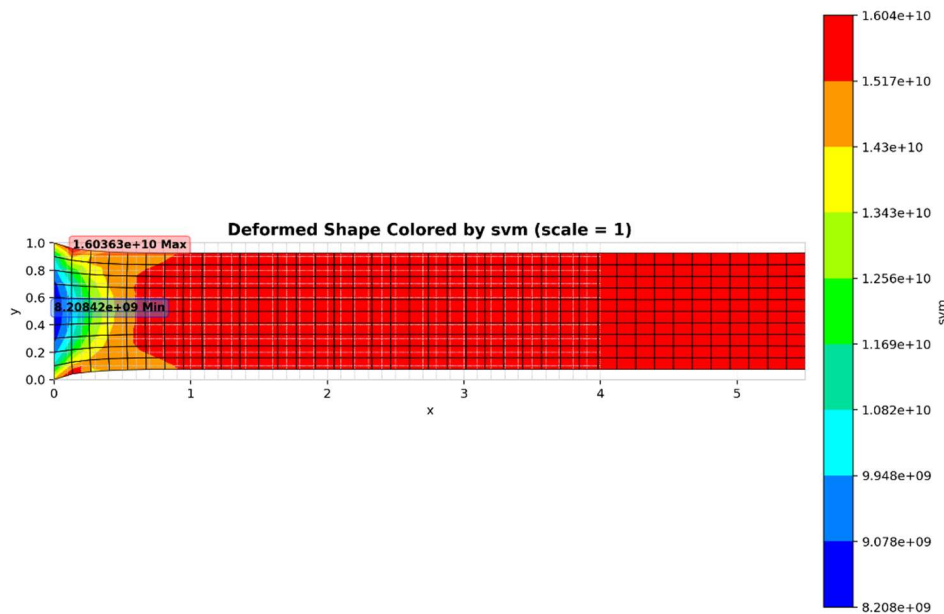


Figure 35. Equivalent von Mises Stress Field, Neo - Hookean material (Program)

Figures 32 to 35 present the von Mises equivalent stress fields for the linear elastic and Neo-Hookean materials, respectively. Among all the response quantities examined, this stress field shows the largest difference between the two material models. In Figure 32 and 34, corresponding to the linear material, the maximum stress values are higher and are mainly concentrated in localized stress-concentration regions near the support. By contrast, in Figure 33 and 35, the Neo-Hookean material exhibits lower maximum stress values, but the stress is distributed over a wider portion of the body, indicating a smoother distribution of the internal forces or over estimation of the max equivalent stress at the stress concentration regions. It is also observed that at the center of the fixed support the stresses are almost zero in both cases. In the linear stress diagram, however, the field shows an isolated stress “island” in the central part of the body. When this pattern is compared with the Neo-Hookean result, it appears likely that this feature is not physically representative and may instead be an artifact of the linear stress prediction under the imposed deformation level. Overall, the comparison suggests that the Neo-Hookean formulation provides a more spatially distributed stress response, whereas the linear model tends to overpredict the peak stresses in localized regions. Therefore, the stress field comparison highlights more clearly than the displacement field the effect of the constitutive law on the structural response.

6.5 Comparison of Load–Displacement Curves

The load–displacement curves shown in Figures 31 and 32 were extracted from the node located at the far down - right corner of the body, where the displacement boundary condition was imposed. Based on the graph, the Neo-Hookean material exhibits a mild relaxing behavior as the loading increases, in the sense that its response departs slightly from the initial tangent and becomes marginally softer than the linear prediction. An analogous prediction is observed in the reactions in the beam’s support, where the Neo-Hookean material’s results show about half a magnitude difference. Overall, the close agreement between the curves obtained from the custom project and from ANSYS further confirms the validity of the proposed implementation and shows that the main findings of the study are consistently reproduced by the commercial benchmark.

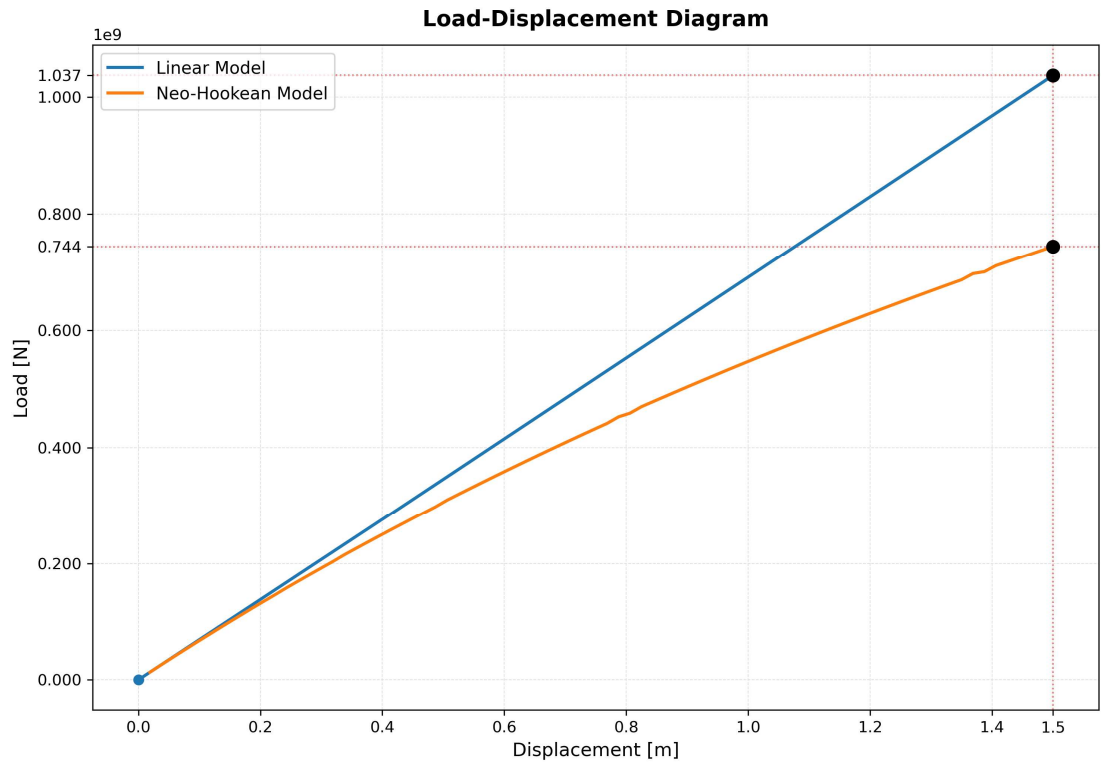


Figure 36.. Load- Displacement diagram, Neo-Hookean Material (Program)

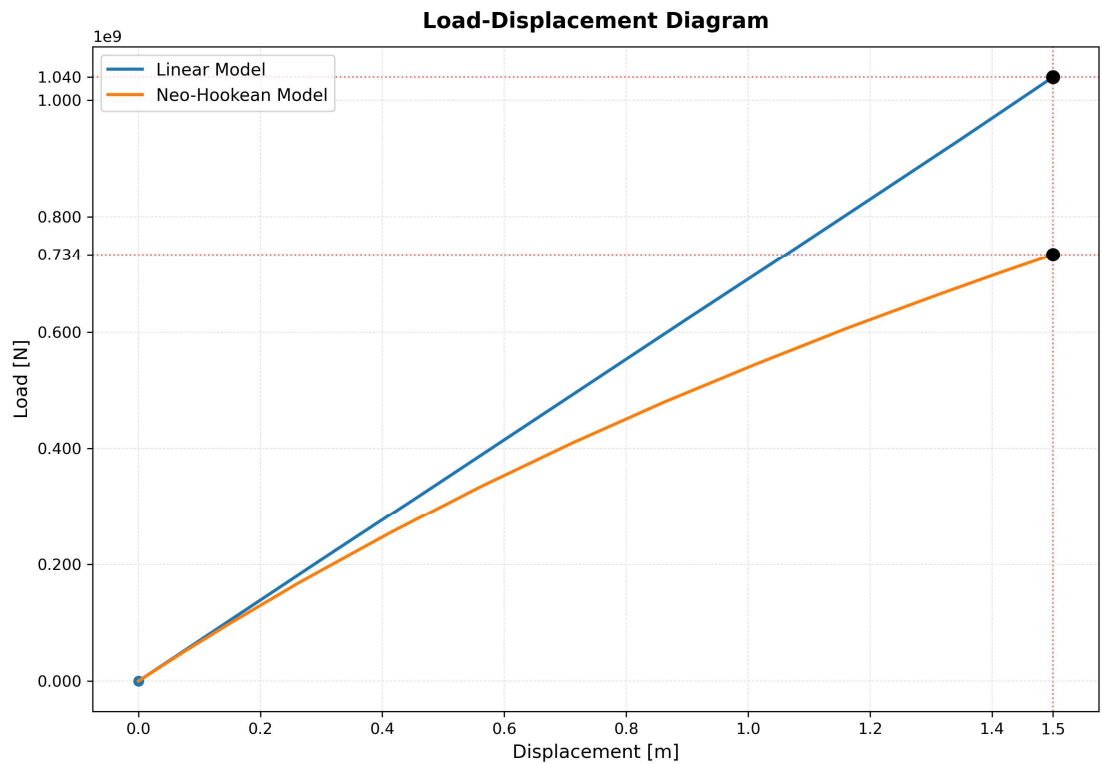


Figure 37. Load -Deflection Diagram (ANSYS)

As shown in Table 8, the linear approach follows a trend similar to that observed in the stress field comparison, namely that it tends to overestimate the support reactions at the fixed boundary. This overprediction is mainly reflected in the horizontal reaction, where the linear

model gives higher values than the Neo-Hookean formulation. By contrast, the vertical reaction remains very close to zero in both cases, as expected from the symmetry of the imposed displacement and the absence of any significant perpendicular loading. It is also observed that the linear approach exhibits slightly better convergence of R toward zero, although the difference is small and does not affect the global interpretation of the results. Overall, the reaction comparison supports the conclusion that the linear material model is slightly stiffer in its global response, whereas the Neo-Hookean formulation leads to a marginally more relaxed redistribution of the support forces.

Table 8. Program - ANSYS Reactions comparison

Approach	Program Rx [GN]	ANSYS Rx [GN]	Program Ry[N]	ANSYS Ry [N]
Neo-Hookean At last step	-14.7	-14.7	$-1.19 \cdot 10^{-7}$	$-1.9 \cdot 10^{-2}$
Linear	-20.7	-20.7	$-1.1 \cdot 10^{-4}$	$-4.8 \cdot 10^{-6}$

Appendix A

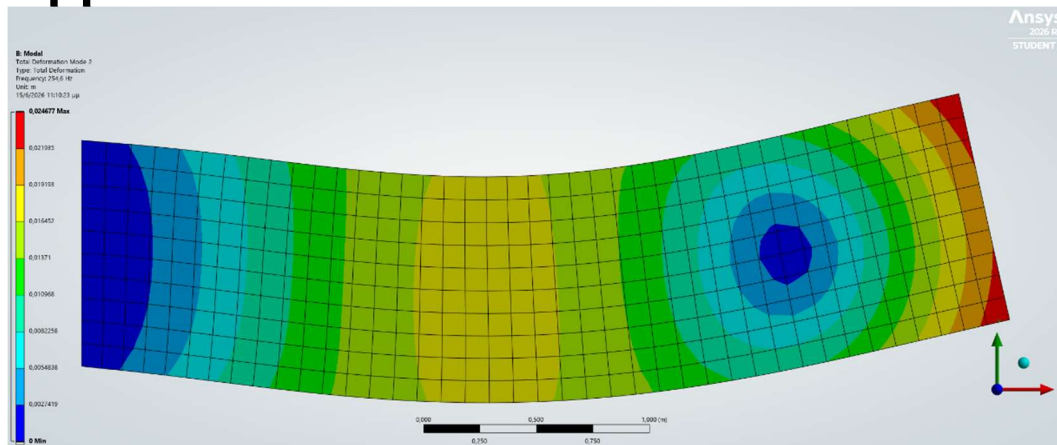


Figure 38. Second mode (ANSYS)

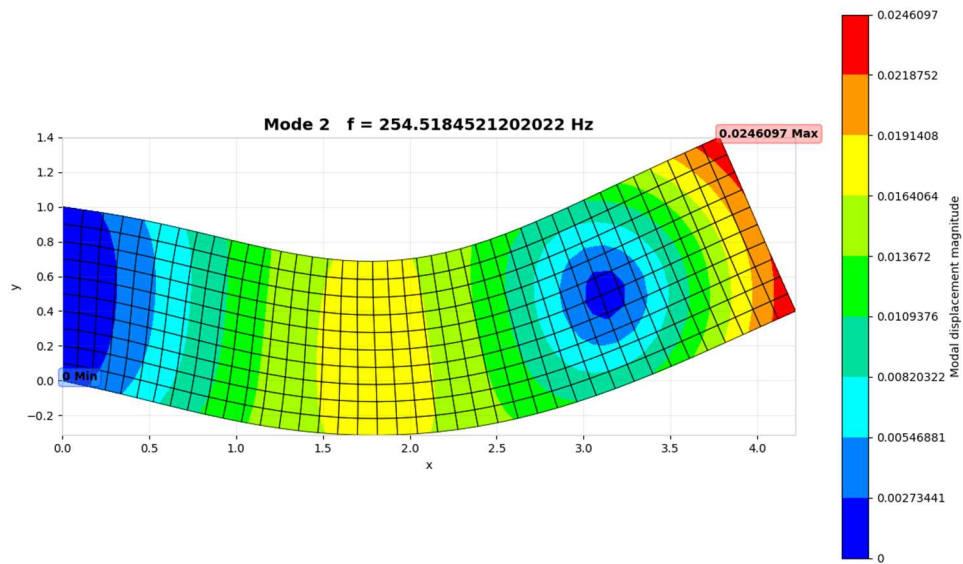


Figure 39. Second mode (Program)

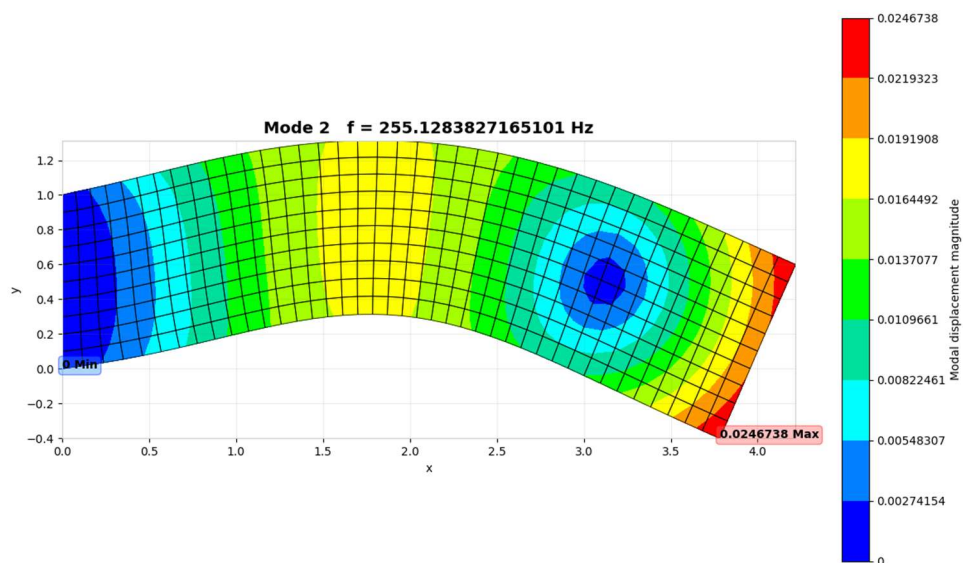


Figure 40. Second mode, Lumped mass assumption (Program)

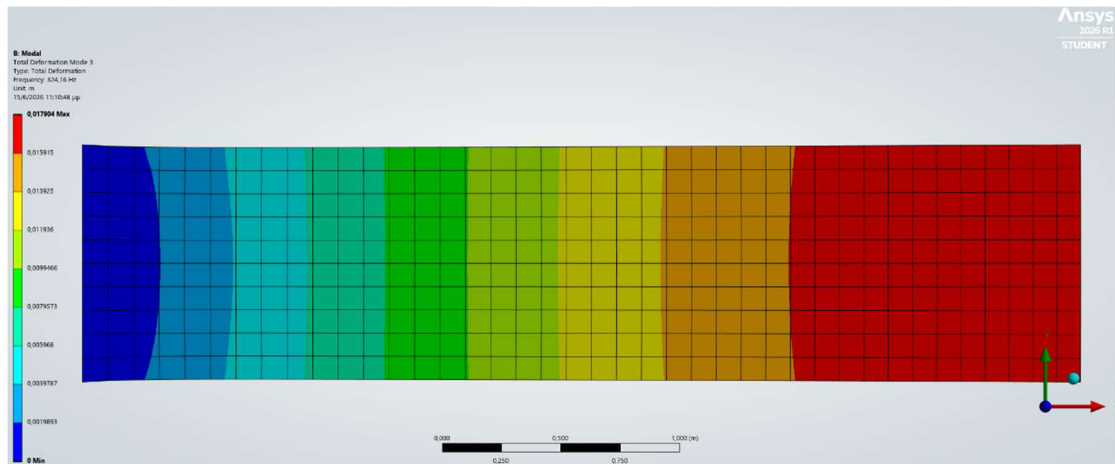


Figure 41. Third mode (ANSYS)

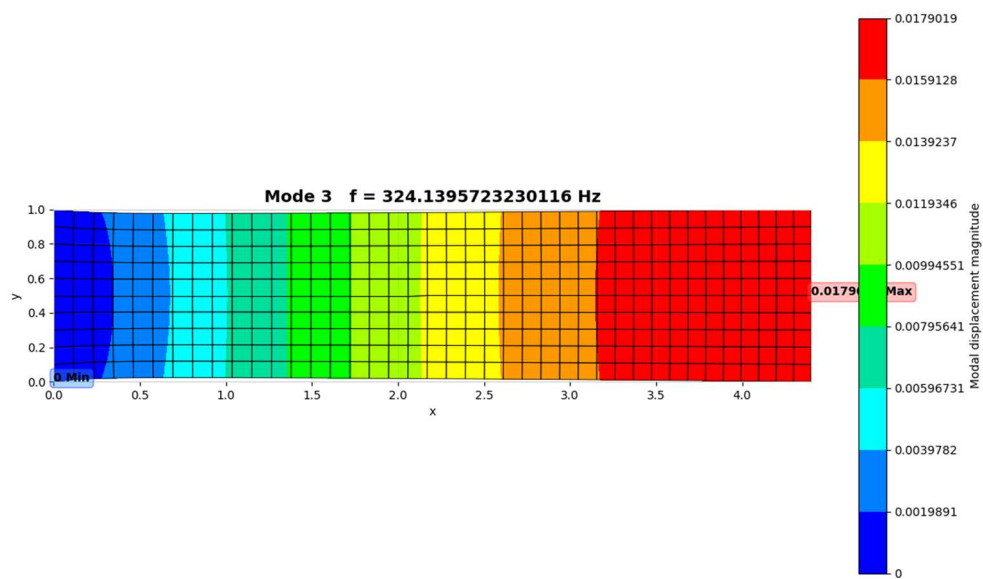


Figure 42. Third mode (Program)

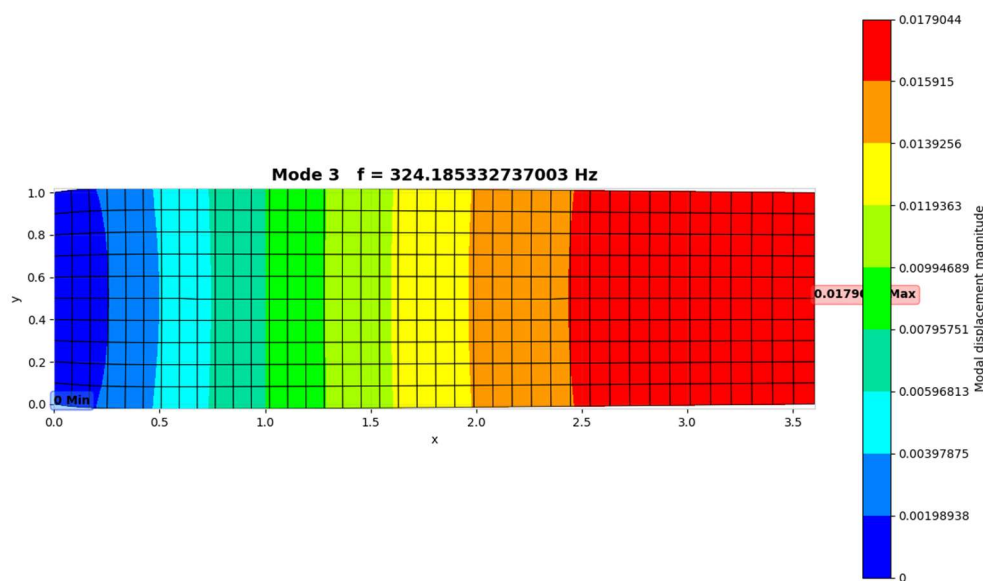


Figure 43. Third mode, Lumped mass assumption (Program)

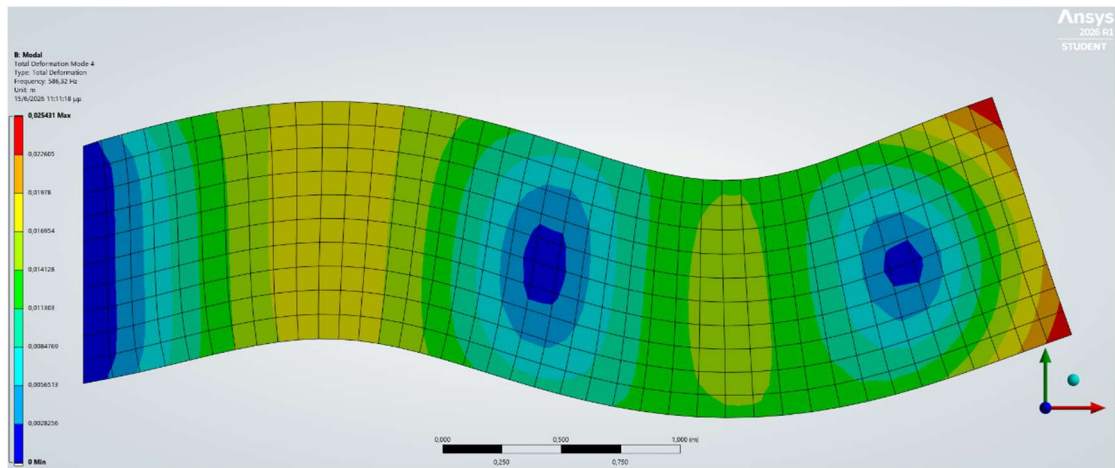


Figure 44. Fourth mode (ANSYS)

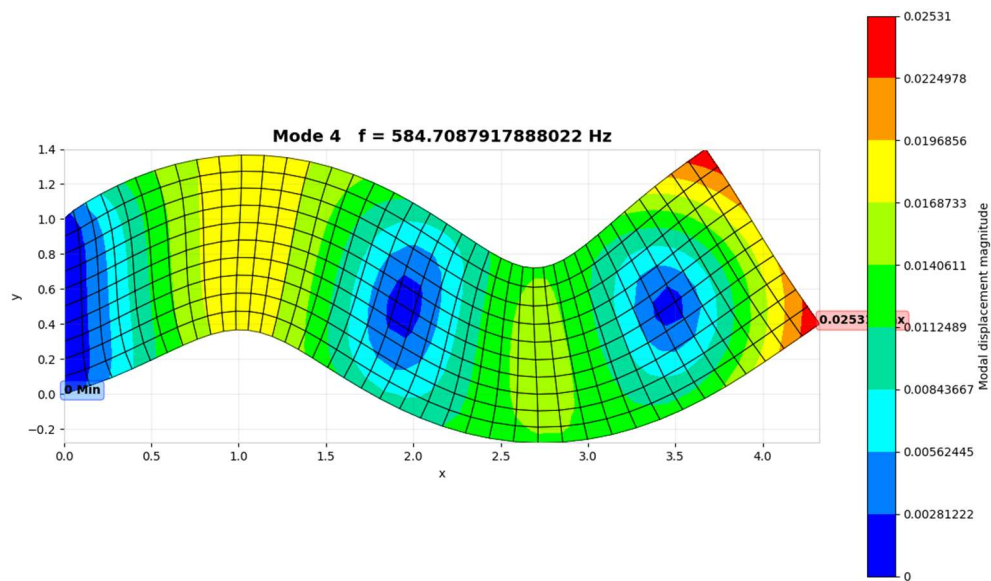


Figure 45. Fourth mode (Program)

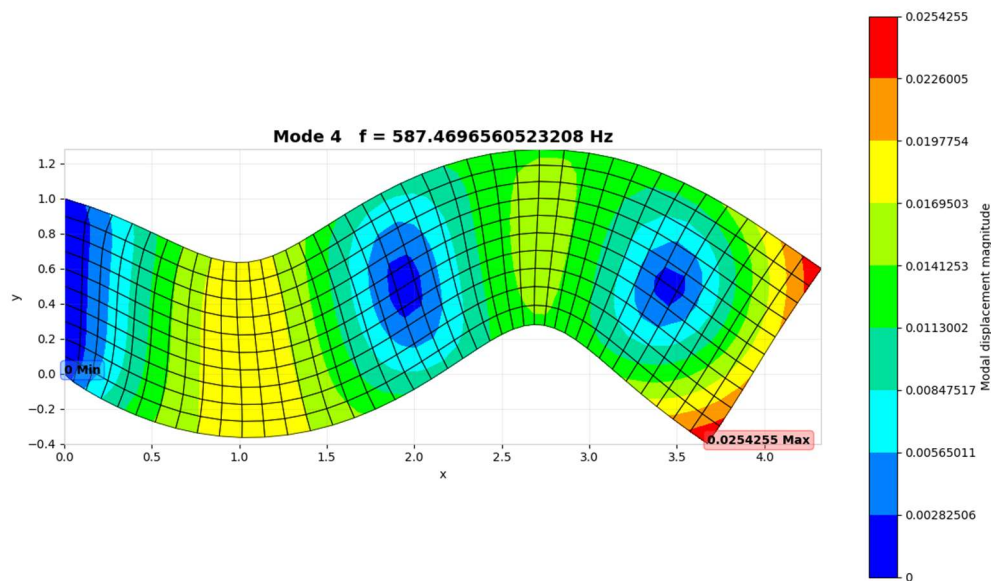


Figure 46. Fourth mode, Lumped mass assumption (Program)